

VirES Aeolus

VirES-Aeolus System Information

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1 Introduction

This document provides details about the Aeolus data presented (Section 2), details about calculating new parameters (Section 3), and the algorithms (Section 4) used in the VirES-Aeolus context.

It shall increase the level of transparency of VirES Aeolus and help users to understand the data operations “hidden” in the background.

2 Aeolus Data Classification

This section defines and describes in detail all Aeolus derived and associated data relevant in the context of VirES-Aeolus. It further provides information about the available parameters of each dataset which are accessible via the VirES-Aeolus interface.

2.1 Data Products

Table 1: Content of Data Products of Aeolus or related to the VirES-Aeolus service

Data Product Type	Content
AISP	Annotated Instrument Source Packets (AISP) - Data downlinked from the satellite for each orbit containing ALADIN spectrometers, instrument and platform status (e.g. settings, temperatures) and AOCS (Attitude and Orbit Control System)
Level 0	Time-ordered source packet (measurement, housekeeping, platform, and AOCS data) streams
Level 1A	Housekeeping source packet fully processed, AOCS source packets (geolocation) processed and assigned to measurement data, measurement data unprocessed
Level 1B	Fully processed, calibrated and georeferenced measurement data including HLOS winds, viewing geometry, ground echo data and product confidence data (PCD)
Level 2A	Aerosol/cloud optical properties, as extinction coefficient, co-polarized backscatter coefficient, and PCD
Level 2B	L2B products represent “consolidated” HLOS wind data and include corrections using actual pressure and temperature information as obtained from numerical models from a NWP centre. Additional corrections are based on retrieved optical properties. Measurements are grouped after a scene classification in cloudy and clear air HLOS winds
Level 2C	L2C products contain two-component wind vector profiles on the location of the Aeolus ground track as obtained after the assimilation process of L2B data products at a ECMWF. L2C data products mainly contain information from the NWP model at ECMWF
Auxiliary	Auxiliary files containing processor settings, instrument characterisation data (including calibration and bias correction), meteorological parameters, and climatology information
Calibration	Auxiliary files containing response calibration results
ADAM	A surface reflectance Database for ESA s Earth Observation Missions

DEM	ACE2 DEM - (Altimeter Corrected Elevation Version 2) Digital Elevation Model with 9 arc-second resolution
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2.2 Data Dimensions

This chapter discusses some aspects for data classification and parameters contained in the different product files w.r.t. its dimension relevant for visualization:

- **1D Data:** This dataset depends on one dimension, e.g. time, geographical position along the orbit; this dataset can be represented as a 1D array or x-y plot using e.g. time or position on the x-axis.
- **2D Data:** In the Aeolus case this data represents a number of vertical profiles and depends on 2 dimensions, e.g. time, geographical position along the orbit and altitude or range; this dataset can be represented as a 2D array or as a colour-coded figure with time on the x-axis, altitude on the y-axis and the value of the dataset in colours.
- **3D Data:** In the Aeolus case this data represents a number of vertical profiles of the 2D vector quantity horizontal wind vector (speed and direction) and depends on 2 dimensions, e.g. time, geographical position along the orbit and altitude or range; this dataset can be represented as an 2D array or arrows, where the length of the arrow is proportional to wind speed and the direction of the arrow is proportional to the wind direction.

1D data are time-ordered sequences of 1 parameter with different temporal resolution depending on the source of the data. Examples of 1D data are:

- platform status and health data (L1B)
- instrument housekeeping data (e.g. temperatures, laser energy) and instrument mode (e.g. wind mode, response calibration, other internal modes) (L1B)
- AOCS data (satellite velocity, attitude angles, LOS pointing direction, satellite velocity on LOS direction) and geolocation data (latitude, longitude) of satellite ground track and measurement track (for one specific range gate, e.g. lowermost range gate) (L1B)
- Ground-return speeds (AUX_ZWC) and wind speed corrections from the harmonic bias estimator (AUX_HBE)
- Solar background intensity from Mie and Rayleigh channel (L1B)
- Atmospheric surface parameters, e.g. wind speed, pressure temperature (AUX_MET)

2D data are time- and range-bin ordered sequences of 1 parameter on measurement and observation time scale, including scene classification and grouping for L2A and L2B products. Examples of 2D data (vertical profiles) are:

- HLOS wind from Mie and Rayleigh channel (L1B, L2B)
- Signal intensity from Mie and Rayleigh channel (L1B, L2B)
- Signal-to-noise ratio (SNR) from Mie and Rayleigh channel (L1B, L2B)
- Scattering ratio from Mie channel (L1B)
- Error-quantifiers from Mie and Rayleigh channel (L1B, L2B)
- Masks (e.g. ground return bins) and PCD for Mie and Rayleigh channel (L1B, L2B)
- Co-polarized aerosol backscatter and extinction coefficient, lidar ratio (L2A)
- Two components of the horizontal wind vector (L2C); those could be considered as two independent parameters (u and v-component, speed and direction) and visualized with 2 independent intensity plots or as arrows with different length and

direction

- Atmospheric parameters, e.g. temperature, wind, specific humidity and cloud parameters (AUX_MET)

3D data are time- and range-bin ordered sequences of a vector with 2 components on grouping and profile scale. Only the L2C data and the AUX_Met file contain a 3D dataset:

- 2 components (u and v, or speed and direction) of the horizontal wind vector (L2C, AUX_MET)

The following additional, static data sources are used for Aeolus and should be visualized as ancillary data:

- ACE2 (Altimeter Corrected Elevation Version 2) model with 9 arcsecond resolution as DEM [WR5], which is also used for the Aeolus End-to-End Simulator and L1B processor

The following additional non-static dataset shall be accessible and visualized in the VirES-Aeolus service:

- Surface albedo in the UV derived as 1-month climatology with 0.1° resolution derived from the ADAM project (A surface reflectance Database for ESA's Earth Observation Missions [WR2]) for nadir and off-nadir pointing. This results in two times 12 global datasets.

2.3 Aeolus Dataset Details

2.3.1 ACE2

Table 2: ACE2 Details

Data Description	ACE2 (Altimeter Corrected Elevation Version 2) is a global Digital Elevation Model (DEM) with a 9 arc-second resolution. It is used by the L1B processor and shall form the terrain-line above which the wind measurements shall be plotted
Documentation	Smith, R. G., and P. A. M. Berry: ACE2: Global Digital Elevation Model. User Guide: http://tethys.eaprs.cse.dmu.ac.uk/ACE2/shared/documentation
Data Source	http://tethys.eaprs.cse.dmu.ac.uk/ACE2/
Update Frequency	Unknown
Data Range	0 – 8848 m (TBC)
Spatial Resolution	Vertical: 10 Meter (TBC) Horizontal: 9-arc-seconds (~300 Meter at the Equator) (TBC)
Temporal Resolution	---

2.3.2 ADAM

Table 3: ADAM Details

Data Description	A surface reflectance (albedo) Database in the UV range (355 nm) for ESAs Earth Observation Missions. The albedo
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	map is derived as monthly climatology with 0.1° resolution derived from the ADAM project.
Documentation	http://adam.noveltis.fr/pdfs/NOV-3895-NT-12403_FR-v2.1.pdf
Data Source	UV values derived in 2 different pointing geometries (nadir, off-nadir) from ADAM surface albedo database http://adam.noveltis.fr/
Update Frequency	no update, static values as climatology
Data Range	0 – 1
Spatial Resolution	0.1 degree pixel size
Temporal Resolution	1-month climatology

2.3.3 AUX_MRC_1B

Table 4: AUX_MRC_1B Details

Data Description	Mie Response Calibration – needed for L1b and L2B processing; essential dataset to derive Mie Wind data
Documentation	[RD6] Section: 7.7; pp. 172
Data Source	ftp://ADDFint@aeolus-ds.eo.esa.int/ADDF/Calibration_Data
Update Frequency	every week
Data Range	
Spatial Resolution	measurements and observations
Temporal Resolution	measurements and observations
Parameter 1 - Frequency_Valid	of structure Frequency_Step_Result as 1D main result and filter
Parameter 2 - Measurement_Response_Valid	of structure Frequency_Step_Result as 1D main result and filter
Parameter 3 - Reference_Pulse_Response_Valid	of structure Frequency_Step_Result as 1D main result and filter
Parameter 4 - Measurement_Response	of structure Frequency_Step_Result as 1D main result and filter
Parameter 5 - Measurement_Error_Mie_Response	of structure Frequency_Step_Result as 1D main result and filter

Parameter 6 - Reference_Pulse_Response	of structure Frequency_Step_Result as 1D main result and filter
Parameter 7 - Reference_Pulse_Error_Mie_Response	of structure Frequency_Step_Result as 1D main result and filter
Parameter 8 - Num_Valid_Measurements	of structure Frequency_Step_Data_Statistics as 1D main result and filter
Parameter 9 - FWHM	of structure Frequency_Step_MC_Results as 1D main result and filter
Parameter 10 - Normalized_Useful_Signal	of structure Frequency_Step_Result as 2D main result
Parameter 11 - Mie_Scattering_Ratio	of structure Frequency_Step_Result as 2D main result
Parameter 12 - Measurement_Mean_Sensitivity	of structure Measurement_Response_Calibration defines slope of line
Parameter 13 - Measurement_Zero_Frequency	of structure Measurement_Response_Calibration defines offset of line
Parameter 14 - Reference_Pulse_Mean_Sensitivity	of structure Reference_Pulse_Response_Calibration defines slope of line
Parameter 15 - Reference_Pulse_Zero_Frequency	of structure Reference_Pulse_Response_Calibration defines offset of line
Parameter 16 - Satisfied_Min_Valid_Freq_Per_Cal	of structure Calibration_Validity_Indicators as calibration validity result in text box
Parameter 17 - Freq_Offset_Data_Monotonic	of structure Calibration_Validity_Indicators as calibration validity result in text box
Parameter 18 - Num_Valid_Frequency_Steps	of structure Calibration_Validity_Indicators as calibration validity result in text box
Parameter 20 - Mean_Sensitivity_Valid	of structure Measurement_Calibration_Validity as calibration validity result in text box
Parameter 21 - Error_Response_Std_Dev_Valid	of structure Measurement_Calibration_Validity as calibration validity result in text box

Parameter 22 - Zero_Freq_Response_Valid	of structure Measurement_Calibration_Validity as calibration validity result in text box
Parameter 23 - Data_Monotonic	of structure Measurement_Calibration_Validity as calibration validity result in text box
Parameter 24 - Mean_Sensitivity_Valid	of structure Reference_Pulse_Calibration_Validity as calibration validity result in text box
Parameter 25 - Error_Response_Std_Dev_Valid	of structure Reference_Pulse_Calibration_Validity as calibration validity result in text box
Parameter 26 - Zero_Freq_Response_Valid	of structure Reference_Pulse_Calibration_Validity as calibration validity result in text box
Parameter 27 - Data_Monotonic	of structure Reference_Pulse_Calibration_Validity as calibration validity result in text box
Parameter 28 - Latitude_of_DEM_Intersection	of structure Frequency_Step_Geolocation as geolocation
Parameter 29 - Longitude_of_DEM_Intersection	of structure Frequency_Step_Geolocation as geolocation

2.3.4 AUX_RRC_1B

Table 5: AUX_RRC_1B

Data Description	Rayleigh Response Calibration – needed for L1B processing and generation of AUX_RBC_L2 (Bias Correction Results - needed for L2B processing); essential dataset to derive Rayleigh Wind data
Documentation	[RD6] Section: 7.8; pp. 183
Data Source	ftp://ADDFint@aeolus-ds.eo.esa.int/ADDF/Calibration_Data
Update Frequency	every week
Data Range	
Spatial Resolution	measurements and observations
Temporal Resolution	measurements and observations
Parameter 1 - Frequency_Valid	of structure Frequency_Step_Result as 1D main result and filter
Parameter 2 - Ground_Frequency_Valid	of structure Frequency_Step_Result as 1D main result and filter

Parameter 3 - Measurement_Response_Valid	of structure Frequency_Step_Result as 1D main result and filter
Parameter 4 - Ground_Measurement_Response_Valid	of structure Frequency_Step_Result as 1D main result and filter
Parameter 5 - Reference_Pulse_Response_Valid	of structure Frequency_Step_Result as 1D main result and filter
Parameter 6 - Measurement_Response	of structure Frequency_Step_Result as 1D main result and filter
Parameter 7 - Measurement_Error_Rayleigh_Response	of structure Frequency_Step_Result as 1D main result and filter
Parameter 8 - Ground_Measurement_Response	of structure Frequency_Step_Result as 1D main result and filter
Parameter 9 - Ground_Measurement_Error_Rayleigh_Response	of structure Frequency_Step_Result as 1D main result and filter
Parameter 10 - Reference_Pulse_Response	of structure Frequency_Step_Result as 1D main result and filter
Parameter 11 - Reference_Pulse_Error_Rayleigh_Response	of structure Frequency_Step_Result as 1D main result and filter
Parameter 12 - Normalized_Useful_Signal	of structure Frequency_Step_Result as 2D main result
Parameter 13 - Num_Valid_Measurements	of structure Frequency_Step_Data_Statistics as 1D main result and filter
Parameter 14 - Ray_Spectrometer_Temp_9	of structure Etalon_Average_Temperature as 1D main result and filter
Parameter 15 - Ray_Spectrometer_Temp_10	of structure Etalon_Average_Temperature as 1D main result and filter
Parameter 16 - Ray_Spectrometer_Temp_11	of structure Etalon_Average_Temperature as 1D main result and filter
Parameter 17 - Thermocouple_8_Ray	of structure RSPT_Average_Temperature as 1D main result and filter

Spectrometer_Thermal_Hood_1	
Parameter 18 - Thermocouple_9_Ray_Spectrometer_Thermal_Hood_2	of structure RSPT_Average_Temperature as 1D main result and filter
Parameter 19 - Thermocouple_10_Ray_Spectrometer_Thermal_Hood_3	of structure RSPT_Average_Temperature as 1D main result and filter
Parameter 20 - Thermocouple_11_Ray_Spectrometer_Thermal_Hood_4	of structure RSPT_Average_Temperature as 1D main result and filter
Parameter 21 - Optical_Baseplate_Average	of structure Frequency_Step_Temperature as 1D main result and filter
Parameter 22 - Altitude	of structure Frequency_Step_Geolocation as 2D main output
Parameter 23 - Latitude_of_DEM_Intersection	of structure Frequency_Step_Geolocation as geolocation
Parameter 24 - Longitude_of_DEM_Intersection	of structure Frequency_Step_Geolocation as geolocation
Parameter 25 - Satisfied_Min_Valid_Freq_Per_Cal	of structure Calibration_Validity_Indicators as calibration validity result in text box
Parameter 26 - Satisfied_Min_Valid_Ground_Freq_Per_Cal	of structure Calibration_Validity_Indicators as calibration validity result in text box
Parameter 27 - Freq_Offset_Data_Monotonic	of structure Calibration_Validity_Indicators as calibration validity result in text box
Parameter 28 - Num_Valid_Frequency_Steps	of structure Calibration_Validity_Indicators as calibration validity result in text box
Parameter 29 - Num_Valid_Ground_Frequency_Steps	of structure Calibration_Validity_Indicators as calibration validity result in text box
Parameter 30 - Mean_Sensitivity_Valid	of structure Measurement_Calibration_Validity as calibration validity result in text box
Parameter 31 - Error_Response_Std_Dev_Valid	of structure Measurement_Calibration_Validity as calibration validity result in text box

Parameter 32 - Zero_Freq_Response_Valid	of structure Measurement_Calibration_Validity as calibration validity result in text box
Parameter 33 - Data_Monotonic	of structure Measurement_Calibration_Validity as calibration validity result in text box
Parameter 34 - Mean_Sensitivity_Valid	of structure Ground_Measurement_Calibration_Validity as calibration validity result in text box
Parameter 35 - Error_Response_Std_Dev_Valid	of structure Ground_Measurement_Calibration_Validity as calibration validity result in text box
Parameter 36 - Zero_Freq_Response_Valid	of structure Ground_Measurement_Calibration_Validity as calibration validity result in text box
Parameter 37 - Data_Monotonic	of structure Ground_Measurement_Calibration_Validity as calibration validity result in text box
Parameter 38 - Mean_Sensitivity_Valid	of structure Reference_Pulse_Calibration_Validity as calibration validity result in text box
Parameter 39 - Error_Response_Std_Dev_Valid	of structure Reference_Pulse_Calibration_Validity as calibration validity result in text box
Parameter 40 - Zero_Freq_Response_Valid	of structure Reference_Pulse_Calibration_Validity as calibration validity result in text box
Parameter 41 - Data_Monotonic	of structure Reference_Pulse_Calibration_Validity as calibration validity result in text box
Parameter 42 - Measurement_Mean_Sensitivity	of structure Measurement_Response_Calibration defines slope of line
Parameter 43 - Measurement_Zero_Frequency	of structure Measurement_Response_Calibration defines offset of line
Parameter 44 - Ground_Measurement_Mean_Sensitivity	of structure Ground_Measurement_Response_Calibration defines slope of line
Parameter 45 - Ground_Measurement_Zero_Frequency	of structure Ground_Measurement_Response_Calibration defines offset of line
Parameter 46 - Reference_Pulse_Mean_Sensitivity	of structure Reference_Pulse_Response_Calibration defines slope of line
Parameter 47 - Reference_Pulse_Zero_Frequency	of structure Reference_Pulse_Response_Calibration defines offset of line

Parameter 48 - List_of_Measurement_ Error_Fit_Coefficients	of structure Measurement_Response_Calibration defines coefficients of polynomial fit
Parameter 49 - List_of_Ground_Measu rement_Error_Fit_Coefficients	of structure Ground_Measurement_Response_Calibration defines coefficients of polynomial fit
Parameter 50 - List_of_Reference_Pul se_Error_Fit_Coefficie nts	of structure Reference_Pulse_Response_Calibration defines coefficients of polynomial fit

2.3.5 AUX_ISR_1B

Table 6: AUX_ISR_1B Details

Data Description	Instrument Spectral Registration - Only internal signal; similar to MRC_1B/RRC_1B; AUX_ISR_1B and AUX_RRC_1B needed to produce AUX_RBC_L2 (needed for L2B processing)
Documentation	[RD6] Section: 7.2; pp. 141
Data Source	ftp://ADDFint@aeolus-ds.eo.esa.int/ADDF/Calibration_Data
Update Frequency	not regular, instrument mode is commanded on-request
Data Range	
Spatial Resolution	measurements and observations
Temporal Resolution	measurements and observations
Parameter 1 - Mie_Valid	of structure ISR_Result as 1D main result and filter
Parameter 2 - Rayleigh_Valid	of structure ISR_Result as 1D main result and filter
Parameter 3 - Fizeau_Transmission	of structure ISR_Result as 1D main result and filter
Parameter 4 - Mie_Response	of structure ISR_Result as 1D main result and filter
Parameter 5 - Rayleigh_A_Response	of structure ISR_Result as 1D main result and filter
Parameter 6 - Rayleigh_B_Response	of structure ISR_Result as 1D main result and filter
Parameter 7 - Num_Raw_Data	of structure Data_Stat as 1D main result and filter
Parameter 8 - Num_Mie_Used	of structure Data_Stat as 1D main result and filter

Parameter 9 - Num_Rayleigh_Used	of structure Data_Stat as 1D main result and filter
Parameter 10 - Accumulated_Laser_E nergy_Mie	of structure Data_Quality as 1D main result and filter
Parameter 11 - Mean_Laser_Energy_M ie	of structure Data_Quality as 1D main result and filter
Parameter 12 - Accumulated_Laser_E nergy_Rayleigh	of structure Data_Quality as 1D main result and filter
Parameter 13 - Mean_Laser_Energy_R ayleigh	of structure Data_Quality as 1D main result and filter
Parameter 14 - Laser_Energy_Drift	of structure Data_Quality as 1D main result and filter
Parameter 15 - Fwhm	of structure Mie_Core_2 as 1D main result and filter
Parameter 16 - Num_Iterations_Core_ 2	of structure Mie_Core_2 as 1D main result and filter
Parameter 17 - Simplex_Quality_Flag	of structure Mie_Core_2 as 1D main result and filter
Parameter 18 - Ray_Spectrometer_Te mp_9	of structure Etalon_Average_Temperature as 1D main result and filter
Parameter 19 - Ray_Spectrometer_Te mp_10	of structure Etalon_Average_Temperature as 1D main result and filter
Parameter 20 - Ray_Spectrometer_Te mp_11	of structure Etalon_Average_Temperature as 1D main result and filter
Parameter 21 - Thermocouple_8_Ray_ Spectrometer_Thermal _Hood_1	of structure RSPT_Average_Temperature as 1D main result and filter
Parameter 22 - Thermocouple_9_Ray_ Spectrometer_Thermal _Hood_2	of structure RSPT_Average_Temperature as 1D main result and filter
Parameter 23 - Thermocouple_10_Ray_ Spectrometer_Thermal _Hood_3	of structure RSPT_Average_Temperature as 1D main result and filter

Parameter 24 - Thermocouple_11_Ray_Spectrometer_Thermal_Hood_4	of structure RSPT_Average_Temperature as 1D main result and filter
Parameter 25 - Optical_Baseplate_Average	of structure ISR_Result as 1D main result and filter
Parameter 26 - Freq_Rayleigh_Filter_Centre	of structure Data_Set_Record as ISR main result in text box
Parameter 27 - Freq_Mie_USR_Closest_to_Rayleigh_Filter_Centre	of structure Data_Set_Record as ISR main result in text box
Parameter 28 - Num_Valid_Mie_Results	of structure Data_Set_Record as ISR main result in text box
Parameter 29 - Num_Valid_Rayleigh_Results	of structure Data_Set_Record as ISR main result in text box

2.3.6 AUX_ZWC_1B

Table 7: AUX_ZWC_1B Details

Data Description	Zero Wind Calibration: Ground-return speeds (AUX_ZWC_1B) and wind speed corrections from the Harmonic Bias Estimator (AUX_HBE_1B); AUX_ZWC_1B is needed by the HBE tool, which in turn delivers input needed by L1B and L2B processing; essential to derive ground velocity corrected winds without AUX_ZWC_1B
Documentation	[RD6] Section: 7.11; pp. 199
Data Source	ftp://ADDFint@aeolus-ds.eo.esa.int/ADDF/Calibration_Data
Update Frequency	
Data Range	
Spatial Resolution	
Temporal Resolution	
Parameter 1 - Latitude_of_DEM_Intersection	of structure Observation_Info as geolocation
Parameter 2 - Longitude_of_DEM_Intersection	of structure Observation_Info as geolocation

Parameter 3 - Mie_Satellite_Range_to_Target	of structure Observation_Info as main 2D data
Parameter 4 - Rayleigh_Satellite_Range_to_Target	of structure Observation_Info as main 2D data
Parameter 5 - ZWC_Result_Type	of structure Data_Set_Record as main 1D data and filter
Parameter 6 - Mie_Ground_Correction_Velocity	of structure Data_Set_Record as main data record 1D data
Parameter 7 - Rayleigh_Ground_Correction_Velocity	of structure Data_Set_Record as main data record 1D data
Parameter 8 - Number_of_Mie_Ground_Bins	of structure Validity_Indicators as main 1D data and filter
Parameter 9 - Number_of_Rayleigh_Ground_Bins	of structure Validity_Indicators as main 1D data and filter
Parameter 10 - Mie_Average_Ground_Echo_Bin_Thickness	of structure Validity_Indicators as main 1D data and filter
Parameter 11 - Rayleigh_Average_Ground_Echo_Bin_Thickness	of structure Validity_Indicators as main 1D data and filter
Parameter 12 - Mie_Average_Ground_Echo_Bin_Thickness_Above_DEM	of structure Validity_Indicators as main 1D data and filter
Parameter 13 - Rayleigh_Average_Ground_Echo_Bin_Thickness_Above_DEM	of structure Validity_Indicators as main 1D data and filter
Parameter 14 - Mie_Min_Top_Ground_Bin	of structure Validity_Indicators as main 1D data and filter
Parameter 15 - Rayleigh_Min_Top_Ground_Bin	of structure Validity_Indicators as main 1D data and filter

Parameter 16 - Mie_Max_Bottom_Ground_Bin	of structure Validity_Indicators as main 1D data and filter
Parameter 17 - Rayleigh_Max_Bottom_Ground_Bin	of structure Validity_Indicators as main 1D data and filter

2.3.7 ALD_U_N_1B (L1B)

Table 8: ALD_U_N_2B Details (L1B)

Data Description	Fully processed, calibrated and georeferenced measurement data including HLOS winds, viewing geometry, ground echo data and product confidence data (PCD)
Documentation	[RD10], and [RD6] section 5 and 6
Data Source	ftp://ADDFint@aeolus-ds.eo.esa.int/ADDF/L1B_Products
Update Frequency	every orbit (≈ 90 min.), except for blind orbits
Data Range	
Spatial Resolution	measurements and observations
Temporal Resolution	measurements and observations
Parameter List	Parameter list according to Table 13

2.3.8 ALD_U_N_2A (L2A)

Table 9: ALD_U_N_2A Details (L2A)

Data Description	Aerosol/cloud optical properties, as extinction coefficient, co-polarized backscatter coefficient, and PCD
Documentation	[RD11], and [RD8] section 3
Data Source	ftp://ADDFint@aeolus-ds.eo.esa.int/ADDF/L2A_Products
Update Frequency	after availability of L1B and AUX_MET data
Data Range	
Spatial Resolution	accumulated groups
Temporal Resolution	accumulated groups
Parameter List	Parameter list according to Table 14

2.3.9 ALD_U_N_2B (L2B)

Table 10: ALD_U_N_2B Details (L2B)

Data Description	L2B products represent “consolidated” HLOS wind data and include corrections using actual pressure and temperature information as obtained from numerical models from a NWP centre. Additional corrections are based on retrieved optical properties. Measurements are grouped after a scene classification in cloudy and clear air HLOS winds
Documentation	[RD12], and [RD9] Section 4
Data Source	ftp://ADDFint@aeolus-ds.eo.esa.int/ADDF/L2B_Products
Update Frequency	after availability of L1B product
Data Range	
Spatial Resolution	accumulated groups; different for clear-air and cloudy scenes
Temporal Resolution	accumulated groups; different for clear-air and cloudy scenes
Parameter List	Parameter list according to Table 15

2.3.10 ALD_U_N_2C (L2C)

Table 11: ALD_U_N_2C Details (L2C)

Data Description	L2C products contain two-component wind vector profiles on the location of the Aeolus ground track as obtained after the assimilation process of L2B data products at a ECMWF. L2C data products mainly contain information from the NWP model at ECMWF
Documentation	[RD12], and [RD9] Section 4
Data Source	ftp://ADDFint@aeolus-ds.eo.esa.int/ADDF/L2C_Products
Update Frequency	every 12 hours
Data Range	
Spatial Resolution	accumulated groups
Temporal Resolution	accumulated groups
Parameter List	Parameter list according to Table 16

2.3.11 AUX_MET

Table 12: AUX_Met Details

Data Description	Profiles of atmospheric parameters, e.g. temperature, wind, specific humidity and cloud parameters from ECMWF along Aeolus orbit (nadir, off-nadir)
Documentation	[RD12], and [RD9] Section 5
Data Source	ftp://ADDFint@aeolus-ds.eo.esa.int/ADDF/Auxiliary_Data/AUX_Met_12
Update Frequency	every 12 hours
Data Range	
Spatial Resolution	profiles every 3 s along track
Temporal Resolution	profiles every 3 s along track
Parameter List	Parameter list according to Table 17

2.4 Aeolus Product Level Information

The following Tables contain information about the specific parameters of the L1B, L2A, L2B, L2C and AUX_MET files, which should be visualized in Globe and Analytics View. The columns describe the parameter name, if data is available from Rayleigh (R) and/or Mie (M) channel, the physical unit (a.u. arbitrary unit in case of unit-less parameter), the dimension, the spatial/temporal resolution or granularity (o: observation, m: measurement, w: wind-accumulation-result (L2B/C), p: profile (AUX_MET)), the corresponding IODD section, the IODD Table, and remark.

2.4.1 ALD_U_N_1B (L1B)

Table 13: List of Parameters from L1b product from Wind Measurement Mode

Parameter Name	Ray / Mie	Unit	Dim.	Gran.	L1B IODD Section	Reference Table	Remark
HLOS wind speed	R/M	m/s	2D	o/m	6.3.8	6-63, 6-64	
SNR	R/M	1	2D	o/m	6.3.2	6-5, 6-8	
Signal intensity	R/M	a.u.	2D	o/m	6.3.7	6-58, 6-59	
Scattering Ratio	M	1	2D	o/m	6.3.2	6-5, 6-8	
Wind error	R/M	m/s	2D	o/m	6.3.2	6-5	

Quality Flag Wind	R/M	flag	2D	o/m	6.3.8	6-63, 6-64	
Time	R/M	UTC	1D	o/m	5.3.1	5-4, 5-5	Centroid time
Latitude	R/M	°N	2D	o/m	5.3.1	5-7, 5-8, 5-9	
Longitude	R/M	°E	2D	o/m	5.3.1	5-7, 5-8, 5-9	
Range	R/M	m	2D	o/m	5.3.1	5-7, 5-8, 5-9	Different for R/M
Altitude	R/M	m	2D	o/m	5.3.1	5-7, 5-8, 5-9	Different for R/M
Instrument Mode		flag	1D	o	5.3.1	5-3	
Platform/ instrument status		flag	1D				tbd
Instrument temperatures		°C	1D				only in L1A and for non-wind modes in L1B
Laser energy		mJ	1D	o/m	6.3.1	6-4, 6-7	
Laser frequency	R/M	GHz	1D	o/m	6.3.1	6-4, 6-7	
Solar background	R/M	a.u.	1D	o/m	6.3.7	6-58, 6-59	atmospheric sample no. 25
Ground return speed	R/M	m/s	1D	o	6.3.8	6-62	
Altitude of DEM inter.		m	1D	o/m	5.3.1	5-6, 5-10	
Lat/Lon of DEM intersection		°N °E	1D	o/m	5.3.1	5-6, 5-10	
Velocity at DEM intersection		m/s	1D	o/m	5.3.1	5-6,5-10	

A OCS data		°	1D	o/m	5.3.1	5-4, 5-5,	Pitch, roll, heading
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Section 6 describes L1B content, but Geolocation in L1B is a copy of the L1A, so corresponding tables are found in Section 5.3.1 from the L1A product.

Reference [RD6]: L1B IODD ICD, Issue 4/06; 1 June 2016 ADM-IC-52-1666

2.4.2 ALD_U_N_2A (L2A)

Table 14: List of Parameters from L2A product

Parameter Name	Ray/Mie	Unit	Dim.	Gran.	L2B IODD Section	Reference Table	Remark
SCA extinction	R	10^{-6} m^{-1}	2D	o	3.5.8	3-32, 3-33	Lat, lon, alt sect. 3.5.1
SCA backscatter	R	$10^{-6} \text{ m}^{-1} \text{ sr}^{-1}$	2D	o	3.5.8	3-32, 3-33	Lat, lon, alt sect. 3.5.1
SCA LOD	R	a.u.	2D	o	3.5.8	3-32, 3-33	Lat, lon, alt sect. 3.5.1
SCA SR	R	a.u.	2D	o	3.5.8	3-32, 3-33	Lat, lon, alt sect. 3.5.1
SCA mid extinction	R	10^{-6} m^{-1}	2D	o	3.5.8	3-32, 3-35, 3-34	Lat, lon, alt in table 3-34
SCA mid backscatter	R	$10^{-6} \text{ m}^{-1} \text{ sr}^{-1}$	2D	o	3.5.8	3-32, 3-35, 3-34	Lat, lon, alt in table 3-34
SCA mid LOD	R	a.u.	2D	o	3.5.8	3-32, 3-35, 3-34	Lat, lon, alt in table 3-34
SCA mid SR	R	a.u.	2D	o	3.5.8	3-32, 3-35, 3-34	Lat, lon, alt in table 3-34
Filling case	R	--	2D	o	3.5.9	3-36, 3-37	Lat, lon, alt sect. 3.5.1

ICA extinction	R	$10^{-6} \cdot \text{m}^{-1}$	2D	o	3.5.8	3-32, 3-33	Lat, lon, alt sect. 3.5.1
ICA backscatter	R	$10^{-6} \cdot \text{m}^{-1} \cdot \text{sr}^{-1}$	2D	o	3.5.8	3-32, 3-33	Lat, lon, alt sect. 3.5.1
ICA LOD	R	a.u.	2D	o	3.5.8	3-32, 3-33	Lat, lon, alt sect. 3.5.1
MCA clim. BER	M	sr^{-1}	2D	o	3.5.10	3-38, 3-39	Lat, lon, alt sect. 3.5.1
MCA extinction	M	$10^{-6} \cdot \text{m}^{-1}$	2D	o	3.5.8	3-38, 3-39	Lat, lon, alt sect. 3.5.1
MCA LOD	M	a.u.	2D	o	3.5.8	3-38, 3-39	Lat, lon, alt sect. 3.5.1
Group extinction	R	$10^{-6} \cdot \text{m}^{-1}$	1D	g	3.5.12	3-42, 3-43	Lat, lon, alt sect. 3.5.1
Group backscatter	R	$10^{-6} \cdot \text{m}^{-1} \cdot \text{sr}^{-1}$	1D	g	3.5.12	3-42, 3-43	Lat, lon, alt sect. 3.5.1
Group LOD	R	a.u.	1D	g	3.5.12	3-42, 3-43	Lat, lon, alt sect. 3.5.1
Group SR	R	a.u.	1D	g	3.5.12	3-42, 3-43	Lat, lon, alt sect. 3.5.1
Group mid extinction top	R	$10^{-6} \cdot \text{m}^{-1}$	1D	g	3.5.12	3-42, 3-45, 3-44	Lat, lon, alt in table 3-44
Group mid backscatter top	R	$10^{-6} \cdot \text{m}^{-1} \cdot \text{sr}^{-1}$	1D	g	3.5.12	3-42, 3-45, 3-44	Lat, lon, alt in table 3-44
Group mid LOD top	R	a.u.	1D	g	3.5.12	3-42, 3-45, 3-44	Lat, lon, alt in table 3-44

Group mid BER top	R	sr^{-1}	1D	g	3.5.12	3-42, 3-45, 3-44	Lat, lon, alt in table 3-44
Group mid extinction bot	R	$10^{-6} * \text{m}^{-1}$	1D	g	3.5.12	3-42, 3-45, 3-44	Lat, lon, alt in table 3-44
Group mid backscatter bot	R	$10^{-6} * \text{m}^{-1} * \text{sr}^{-1}$	1D	g	3.5.12	3-42, 3-45, 3-44	Lat, lon, alt in table 3-44
Group mid LOD bot	R	a.u.	1D	g	3.5.12	3-42, 3-45, 3-44	Lat, lon, alt in table 3-44
Group mid BER bot	R	sr^{-1}	1D	g	3.5.12	3-42, 3-45, 3-44	Lat, lon, alt in table 3-44
Group Aladin cloud flag	R	--	1D	g	3.5.13	3-46	
Group Aladin cloud flag	R	--	1D	g	3.5.13	3-46	
Group reliability	R	--		g	3.5.13	3-46	
ICA First matching bin	R	--	1D	o	3.5.4	3-25	Lat, lon, alt sect. 3.5.1
ICA QC flag	R	--	1D	o	3.5.4	3-25	Lat, lon, alt sect. 3.5.1
SCA extinction variance	R	m^{-2}	2D	o	3.5.3	3-22, 3-23	Lat, lon, alt sect. 3.5.1
SCA backscatter variance	R	$\text{m}^{-2} * \text{sr}^{-2}$	2D	o	3.5.3	3-22, 3-23	Lat, lon, alt sect. 3.5.1
SCA LOD variance	R	a.u.	2D	o	3.5.3	3-22, 3-23	Lat, lon, alt sect. 3.5.1

SCA mid extinction variance	R	m^{-2}	2D	o	3.5.3	3-22, 3-24, 3-34	Lat, lon, alt table 3-34
SCA mid backscatter variance	R	$m^{-2} * sr^{-2}$	2D	o	3.5.3	3-22, 3-24, 3-34	Lat, lon, alt table 3-34
SCA mid LOD variance	R	a.u.	2D	o	3.5.3	3-22, 3-24, 3-34	Lat, lon, alt table 3-34
SCA mid BER variance	R	sr^{-2}	2D	o	3.5.3	3-22, 3-24, 3-34	Lat, lon, alt table 3-34
Group extinction variance	R	m^{-2}	1D	o	3.5.7	3-31	Lat, lon, alt sect. 3.5.1
Group backscatter variance	R	$m^{-2} * sr^{-2}$	1D	o	3.5.7	3-31	Lat, lon, alt sect. 3.5.1
Group LOD variance	R	a.u.	1D	o	3.5.7	3-31	Lat, lon, alt sect. 3.5.1
Group mid extinction variance top	R	m^{-2}	1D	o	3.5.7	3-31, 3-34	Lat, lon, alt table 3-34
Group mid backscatter variance top	R	$m^{-2} * sr^{-2}$	1D	o	3.5.7	3-31, 3-34	Lat, lon, alt table 3-34
Group mid LOD variance top	R	a.u.	1D	o	3.5.7	3-31, 3-34	Lat, lon, alt table 3-34
Group mid BER variance top	R	sr^{-2}	1D	o	3.5.7	3-31, 3-34	Lat, lon, alt table 3-34
Group mid extinction variance bot	R	m^{-2}	1D	o	3.5.7	3-31, 3-34	Lat, lon, alt table 3-34
Group mid backscatter variance bot	R	$m^{-2} * sr^{-2}$	1D	o	3.5.7	3-31, 3-34	Lat, lon, alt table 3-34
Group mid LOD variance bot	R	a.u.	1D	o	3.5.7	3-31, 3-34	Lat, lon, alt table 3-34

Group mid BER variance bot	R	sr ⁻²	1D	o	3.5.7	3-31, 3-34	Lat, lon, alt table 3-34
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Longitude of DEM		10-6degE	1D	o	3.5.1	3-8	
Latitude of DEM		10-6degN	1D	o		3-8	

Reference [RD8]: Aeolus Level 2a Processor Input/Output Data Definition, V3.04, 17.06.2016, AE-IF-DLR-L2A-004.

2.4.3 ALD_U_N_2B (L2B)

Table 15: List of Parameters from L2B product

Parameter Name	Ray / Mie	Unit	Dim.	Gran.	L1B IODD Section	Reference Table	Remark
HLOS wind speed	R/M	m/s	2D	w	4.3.8, 4.3.9	53, 55	Different for Clear and Cloudy, and for R/M
SNR	R/M	a.u.	2D	w/m	4.3.6,	47	
Scattering Ratio	R/M	a.u.	2D	w/m	4.3.5, 4.3.6, 4.3.7	40, 43, 47, 49	
Wind error	R/M	m/s	2D	w	4.3.6, 4.3.7	47, 49	Different for R/M
Quality Flag	R/M	flag	2D	m	4.3.5,	31, 32, 33, 35, 36, 37	Different for R/M
Quality Flag	R/M	flag	2D	w	4.3.6, 4.3.7	47, 49	Different for R/M
Latitude		°N	2D	w	4.3.3	21	
Longitude		°W	2D	w	4.3.3	21	
Range	R/M	m	2D	w	4.3.3	21	Different for R/M
Altitude	R/M	m	2D	w	4.3.3	21	Different for R/M
[Mie/Rayl]_Map_of_L	R/M		2D	m	4.3.1	15,16,17	

1B_Meas_Used							
Grouping definition	R/M			m	4.3.2	18	Different for R/M
Bin classification	R/M	Clear/ cloudy	2D	m	4.3.5	40,43	Different for R/M

extinction	R	1/m	2D	m	4.3.5	44	
cloud/aerosol layer location		m	2D		4.3.5	44	Derived from Rayleigh channel

back.: background

Reference [RD9]: L2B/L2C IODD ICD, V. 2.30, 1 July 2016, AE-IF-ECMWF-L2BP-001

2.4.4 ALD_U_N_2C (L2C)

Table 16: List of Parameters from L2C product

Parameter Name	Ray / Mie	Unit	Dim.	Gran.	L1B IODD Section	Reference Table	Remark
Analysis and back. vector wind	R/M	m/s	3D	w	4.3.13	72, 73, 74	Different for R/M, L2C
analysis and back. HLOS wind	R/M	m/s	2D	w	4.3.11	65	L2C
Screening flags	R	flag	2D	w	4.3.11 4.3.12	63, 70	L2C

back.: background

with information from L2C file one could calculate observation – analysis (o-a) and observation – background (o-b) values and statistics, which is used within the NWP community. This is done by calculating the differences of L2B to L2C product. The screening flag provides information, if the data is assimilated or not.

Reference [RD9]: L2B/L2C IODD ICD, V. 2.30, 1 July 2016, AE-IF-ECMWF-L2BP-001

2.4.5 AUX_MET

Table 17: List of Parameters from AUX_MET File

Parameter Name	Ray / Mie	Unit	Dim.	Gran.	L2B/2C IODD Section	Reference Table	Remark
Latitude		°N	1D	p	5.1.7	79	
Longitude		°E	1D	p	5.1.7	79	
Date and Time		UTC	1D	p	5.1.7	79	
Altitude		m	2D	p	5.1.8	80/81	+ surface value
Pressure p		hPa	2D	p	5.1.8	80/81	+surface value
Temperature T		K	2D	p	5.1.8	81	
Relative humidity RH		%	2D	p	5.1.8	81	
Specific humidity SH		kg/kg	2D	p	5.1.8	81	
Wind component U		m/s	2D	p	5.1.8	80/81	+surface value
Wind component V		m/s	2D	p	5.1.8	80/81	+surface value
Cloud Cover CC		%	2D	p	5.1.8	81	
Cloud liquid water content CLWC		kg/kg	2D	p	5.1.8	81	
Cloud ice water content CIWC		kg/kg	2D	p	5.1.8	81	

column Ray/Mie is not applicable for AUX_MET data

Altitude, wind component U, V and pressure are also available at the surface (Table 80); thus values from Table 80 and 81 should be merged

other parameters are available at pressure levels of the model; the units of the

parameters in the file could be are different than provided in the table, e.g. cm/s for wind speeds, cm for altitudes, and Pa for pressure.

Profile data is provided on predicted orbit tracks every 3 s for nadir and off-nadir LOS pointing direction; off-nadir data should be displayed by default; nadir-data should be displayed as option. The geolocation of the data is on an own horizontal track, aligned to the track but not necessarily on the same geolocation or with the same horizontal resolution as in the L1B/L2B product.

Reference [RD9] L2B/L2C IODD ICD, V. 2.30; 1 July 2016, AE-IF-ECMWF-L2BP-001

3 Calculation of new parameters

This section provides details about the algorithms needed to perform spatio-temporal grid operations and about the co-location of ancillary datasets with Aeolus data products in space and time. Moreover, it describes the calculation of new parameters needed for proper data visualisation, filtering and geo-referencing of certain parameters.

For certain cases, the calculation of new parameters, from existing Aeolus product parameters, provides additional benefit for the user. In particular, the possibilities of data visualisation and data filtering are enhanced. Apart from that, for some Aeolus data products it is necessary to derive new geolocation parameters from existing ones to ensure correct geo-referencing of the measurement data.

In the following sections, the different spatial and temporal resolutions of the various data products are described for clarification at the beginning of each section. Thereafter new parameters for each product are described on the basis of simple mathematical expressions. At the end of each section, tables relate the variables used for the formulas to the product parameters as given in the VDDA Parameter table [RD3]. Constants and parameters not specified in the data product itself but used for the calculations below (e.g. radiometric gain from the Rayleigh detection chain) are collected in a VirES Aeolus Parameter File.

3.1 Level 1B

This data product includes parameters at two different horizontal resolutions: measurement and observation level. Measurements (3 km horizontal resolution) are the result of the accumulation of 20 laser pulses at a repetition rate of 50.5 Hz and the average of 30 measurements is defined as observation (90km horizontal resolution).

The L1B data product includes Rayleigh signal intensities through Rayleigh filters A and B on measurement as well as on observation level. For the sake of usefulness, intensities through filter A and B are added to get new parameters describing total Rayleigh signal intensities on measurement and observation level. Thereby, following new parameters are generated:

- **Rayleigh signal intensity obs. $I_{tot,obs}$:** It is calculated from the sum of signal intensities through Rayleigh channel A $I_{A,obs}$ and Rayleigh channel B $I_{B,obs}$ according to following equation:

$$I_{tot,obs}(j,i) = I_{A,obs}(j,i) + I_{B,obs}(j,i)$$

where j is the observation index and i is the range gate index.

- **Rayleigh signal intensity meas. $I_{tot, meas}$:** In the same way as described above, this parameter is defined as the sum of the parameters $I_{A, meas}$ and $I_{B, meas}$:

$$I_{tot, meas}(j, k, i) = I_{A, meas}(j, k, i) + I_{B, meas}(j, k, i)$$

where j is the observation index, k describes the measurement index and i is the range gate index.

In the same way as for Rayleigh intensities, signal-to-noise values are provided for each channel. Given the parameters of the L1B data product, it is possible to calculate a signal-to-noise value representative for the total Rayleigh intensity. This parameter is calculated using the signal intensities through both filters and the Rayleigh background signal based on the signal-to-noise calculations given in [RD7] (eq. 79.6-79.9). Following new parameters are calculated:

- **Rayleigh SNR meas. $SNR_{tot, meas}$:**

$$SNR_{tot, meas}(j, k, i) = \frac{1}{\sqrt{k_{Ray}} \sqrt{I_{tot, meas}(j, k, i) + 2 * [I_{A, meas}(j, k, i = 25) + I_{B, meas}(j, k, i = 25)]}}$$

where j is the observation index, k the measurement index and i is the range gate index and $I_{A, meas}(j, k, i = 25)$ and $I_{B, meas}(j, k, i = 25)$ describe the background signals for channel A and B on measurement level. The parameter k_{Ray} is the radiometric gain from the Rayleigh detection chain and has a value in the order of $k_{Ray} = 0.56413$ LSB/electron [RD7]. This parameter can change in course of time and is defined in the VirES Aeolus Parameter File.

- **Rayleigh SNR obs. $SNR_{tot, obs}$:**

$$SNR_{tot, obs}(j, i) = \frac{1}{\sqrt{k_{Ray}} \sqrt{I_{tot, obs}(j, i) + 2 * [I_{A, obs}(j, i = 25) + I_{B, obs}(j, i = 25)]}}$$

where j is the observation index and i is the range gate index and $I_{A, obs}(j, i = 25)$ and $I_{B, obs}(j, i = 25)$ describe the background signals for channel A and B on observation level. The parameter k_{Ray} is the radiometric gain from the Rayleigh detection chain and has a value in the order of $k_{Ray} = 0.56413$ LSB/electron [RD7]. This parameter can change in course of time and is defined in the VirES Aeolus Parameter File.

Table 18: Parameters of [RD3] corresponding to formula variables of Section 3.1

Variable	VDDA-PT name
$I_{A,obs}$	Rayleigh signal channel A intensity obs.
$I_{B,obs}$	Rayleigh signal channel B intensity obs.
$I_{tot,obs}$	Rayleigh signal intensity obs.
$I_{A,meas}$	Rayleigh signal channel A intensity meas.
$I_{B,meas}$	Rayleigh signal channel B intensity meas.
$I_{tot,meas}$	Rayleigh signal intensity meas.
$SNR_{tot,obs}$	Rayleigh SNR obs.
$SNR_{tot,meas}$	Rayleigh SNR meas.

3.2 Level 2A

The Level 2A data product contains parameters on L2A group and observation level (same definition as for the L1B data product). The principle of the L2A grouping algorithm is to provide L2A products at a better resolution than the resolution of the L1B observation. Thus, the grouping algorithm identifies blocks of measurements having similar properties within one observation and processes them as single features. The grouping algorithm is restricted to one observation, so one L2A group can never exceed the horizontal length of one observation.

Compared to other data products, time parameters of the L2A data product are given w.r.t the start points of the observation or group. This convention leads to problems when comparing L2A parameters with parameters from other data products (see Section 4.5) where the centroid time of the observation or group is used. Thereby, following new parameters are calculated for the L2A data product:

- **L1B centroid time obs. $T_{centr,obs}$:** It's calculated from the mean of two subsequent start time values $Time_{start,obs}$. Only for the last observation of the imported data, $T_{centr,obs}$ is calculated in a different way. The representation of the time values of the formulas below is a value containing the amount of seconds elapsed since 01-Jan-2000 00:00:00 UTC:

$$\begin{aligned}
 T_{centr,obs}(j) &= \frac{1}{2} [T_{start,obs}(j+1) - T_{start,obs}(j)] \quad \text{for } j = 1 \dots (J-1) \\
 &\quad T_{centr,obs}(j = J) \\
 &= T_{centr,obs}(j = J) + \frac{1}{2} [T_{start,obs}(j = J) - T_{start,obs}(j = J-1)]
 \end{aligned}$$

where j is the observation index and J the total number of observations of the analysed section.

- **Group end time $T_{end,group}$:** Start times of the group are already specified in the L2A data product but in order to specify the end times of the L2A groups, the indices of the observation at the end point of the group $g_{end,obs}$ and the last measurement of the observation at the end point of the group $g_{end,meas}$ are used to derive the group end time from the measurement times $T_{centr,meas}$:

$$T_{end,group}(g) = T_{centr,meas}(j = g_{end,obs}, k = g_{end,meas})$$

where j, k describe the observation and measurement indices and g is the group index.

- **Group centroid time $T_{centr,group}$:** Especially for the co-location of the L2A group results with L2B accumulated results, the centroid time of L2A group results is used. It's calculated from the mean of the corresponding group start time $T_{start,group}$ and group end time $T_{end,group}$:

$$T_{centr,group}(g) = \frac{1}{2}[T_{start,group}(g) + T_{end,group}(g)]$$

where g denotes the group index.

The problem for the visualisation of L2A parameters on observation level is that geolocation data necessary for the data visualisation as the geolocation of the DEM intersection point and altitudes of the range bins is only specified on measurement level. For correct georeferencing of the L2A measurement data on observation level, the geolocation parameters mentioned above need to be translated to observation level. Thereby, following new parameters are generated:

- **Longitude of DEM intersection obs. $Lon_{DEM,obs}$:** This parameter is calculated by comparing the measurement times $T_{centr,meas}$ with the centroid time of the observation $T_{centr,obs}$. The index of $T_{centr,meas}$ closest to $T_{centr,obs}$ is used to select the measurement longitude value $Lon_{DEM,meas}$ representative for the whole observation.
- **Latitude of DEM intersection obs. $Lat_{DEM,obs}$:** This parameter is calculated in the same way as described above using the corresponding measurement value $Lat_{DEM,meas}$, $T_{centr,meas}$ and $T_{centr,obs}$.
- **Altitude of DEM intersection obs. $Alt_{DEM,obs}$:** This parameter is calculated by selecting the maximum altitude of DEM intersection value of the measurements composing one observation:

$$Alt_{DEM,obs}(j) = \max_{k \in K} \{Alt_{DEM,meas}(j, k)\}$$

where j and k describe the observation and measurement indices, K is the number of measurements for the corresponding observation and $Alt_{DEM,meas}$ the altitude of the DEM intersection on measurement level.

- **Mie altitude obs. $Mie_{alt,obs}$:** For each range bin and for each observation the altitude values on measurement level $Mie_{alt,meas}$ are horizontally averaged to derive one value representative for the whole observation:

$$Mie_{alt,obs}(j, i) = \frac{1}{K(j)} \sum_{k=1}^{K(j)} Mie_{alt,meas}(j, k, i)$$

where j , k and i are the observation, measurement and range gate indices and K is the number of measurements for the corresponding observation.

- **Rayleigh altitude obs. $Ray_{alt,obs}$:** This parameter is calculated in the same way as described above for the Mie channel:

$$Ray_{alt,obs}(j, i) = \frac{1}{K(j)} \sum_{k=1}^{K(j)} Ray_{alt,meas}(j, k, i)$$

where j , k and i are the observation, measurement and range gate indices, $Ray_{alt,meas}$ the Rayleigh altitudes for each measurement and range bin and K is the number of measurements for the corresponding observation.

Table 19: Parameters of [RD3] corresponding to formula variables of Section 3.2

Variable	VDDA-PT name
$Lon_{DEM,obs}$	Longitude of DEM intersection obs.
$Lon_{DEM,meas}$	Longitude of DEM intersection meas.
$Lat_{DEM,obs}$	Latitude of DEM intersection obs.
$Lat_{DEM,meas}$	Latitude of DEM intersection meas.
$Alt_{DEM,obs}$	Altitude of DEM intersection obs.
$Alt_{DEM,meas}$	Altitude of DEM intersection meas.
K	L1B num. of measurements per obs.

$Mie_{alt,obs}$	Mie altitude obs.
$Ray_{alt,obs}$	Rayleigh altitude obs.
$Mie_{alt,meas}$	Mie altitude meas.
$Ray_{alt,meas}$	Rayleigh altitude meas.
$T_{start,obs}$	L1B start time obs.
$T_{centr,obs}$	L1B centroid time obs.
$T_{centr,meas}$	L1B time meas.
$T_{start,group}$	Group start time
$T_{centr,group}$	Group centroid time
$T_{end,group}$	Group end time
$g_{end,obs}$	Group end obs.
$g_{end,meas}$	Group end meas./obs.

3.3 Level 2B

The L2B data product contains parameters at measurement (same as L1B measurement), group (different from L2A group) and wind result level. The L2B wind retrieval is based on the accumulation of L1B data. First of all, the profiles of L1B measurements are assigned to groups. The horizontal extent of the groups thereby varies depending on the L2B processor settings and can be longer than the extent of one L1B observation. The next step is the classification of L1B measurements according to their optical properties. In this way, measurements are categorised into “clear” and “cloudy” measurements. The L2B wind retrieval is then performed on the accumulated signal intensities from measurement bins that have the same optical properties within one group. The accumulation is only performed in the horizontal dimension (across measurements). The output of the accumulation are wind results for clear and cloudy cases for both channels. In contrast to L2A groups, the horizontal extent of L2B groups is not restricted to one L1B observation.

The geolocation of these so-called wind results is given by a centre-of-gravity value determined by the geolocations of the L1B data used for the accumulation. Consequently, each L2B wind result is assigned to one value of the geolocation parameters describing the intersection with the DEM. The same is true for the parameters describing the geoid separation values for both channels.

As a result, it occurs that there are multiple values describing the intersection point with the DEM, i.e. latitude/longitude and altitude of the DEM intersection when viewing profiles of L2B wind results. To overcome this problem, the altitude value of the lowest valid L2B range bin is selected to be representative for the whole profile. For the case that a profile

does not contain a valid wind result, no intersection values are calculated. Thereby, following new parameters are calculated:

- **Mie profile longitude of DEM intersection $Mie Lon_{DEM,profile}$:** This parameter is derived using the longitudes of DEM intersection given per wind result $Mie Lon_{DEM,wr}$ and the profile-wind-id matrix p_id_{Mie} ¹ which links the Mie wind results to the range gates of the Mie profiles. For each profile, the longitude of DEM intersection value $Mie Lon_{DEM,wr}$ of the lowest (in terms of altitude) valid range bin is selected:

$$Mie Lon_{DEM,profile}(p) = Mie Lon_{DEM,wr}(p_id_{Mie}(p, i_{max}))$$

where p describes the profile index, i_{max} the range gate index of the last (lowest in terms of altitude) valid Mie wind result.

- **Mie profile latitude of DEM intersection $Mie Lat_{DEM,profile}$:** In the same way as described above, this parameter is calculated:

$$Mie Lat_{DEM,profile}(p) = Mie Lat_{DEM,wr}(p_id_{Mie}(p, i_{max}))$$

where p describes the profile index, i_{max} the range gate index of the last (lowest in terms of altitude) valid Mie wind result, $Mie Lat_{DEM,wr}$ the latitudes of DEM intersection per wind result and p_id_{Mie} the profile-wind-id matrix for the Mie channel.

- **Mie profile altitude of DEM intersection $Mie Alt_{DEM,profile}$:** The same procedure as described above, is used to derive this parameter:

$$Mie Alt_{DEM,profile}(p) = Mie Alt_{DEM,wr}(p_id_{Mie}(p, i_{max}))$$

where p describes the profile index, i_{max} the range gate index of the last (lowest in terms of altitude) valid Mie wind result, $Mie Alt_{DEM,wr}$ the altitudes of DEM intersection per wind result and p_id_{Mie} the profile-wind-id matrix for the Mie channel.

- **Mie profile geoid separation $Mie geoid_{profile}$:** This parameter is calculated in the same way as the previous parameters:

$$Mie geoid_{profile}(p) = Mie geoid_{wr}(p_id_{Mie}(p, i_{max}))$$

¹ Indices of range bins not assigned to an accumulated result are set to zero. Valid results refer to the non-zero indices of a profile.

where p describes the profile index, i_{max} the range gate index of the last (lowest in terms of altitude) valid Mie wind result, $Mie\ geoid_{wr}$ the geoid separation values per wind result and p_id_{Mie} the profile-wind-id matrix for the Mie channel.

- **Rayleigh profile longitude of DEM intersection $Ray\ Lon_{DEM,profile}$:** It's calculated in the same way as for the corresponding Mie Parameter:

$$Ray\ Lon_{DEM,profile}(p) = Ray\ Lon_{DEM,wr}(p_id_{Ray}(p, i_{max}))$$

where p describes the profile index, i_{max} the range gate index of the last (lowest in terms of altitude) valid Rayleigh wind result, $Ray\ Lon_{DEM,wr}$ the longitudes of DEM intersection per wind result and p_id_{Ray} the profile-wind-id matrix for the Rayleigh channel.

- **Rayleigh profile latitude of DEM intersection $Ray\ Lat_{DEM,profile}$:** The same procedure as above is applied:

$$Ray\ Lat_{DEM,profile}(p) = Ray\ Lat_{DEM,wr}(p_id_{Ray}(p, i_{max}))$$

where p describes the profile index, i_{max} the range gate index of the last (lowest in terms of altitude) valid Rayleigh wind result, $Ray\ Lat_{DEM,wr}$ the latitudes of DEM intersection per wind result and p_id_{Ray} the profile-wind-id matrix for the Rayleigh channel.

- **Rayleigh profile altitude of DEM intersection $Ray\ Alt_{DEM,profile}$:** Calculated as shown for the previous parameter:

$$Ray\ Alt_{DEM,profile}(p) = Ray\ Alt_{DEM,wr}(p_id_{Ray}(p, i_{max}))$$

where p describes the profile index, i_{max} the range gate index of the last (lowest in terms of altitude) valid Rayleigh wind result, $Ray\ Alt_{DEM,wr}$ the altitudes of DEM intersection per wind result and p_id_{Ray} the profile-wind-id matrix for the Rayleigh channel.

- **Rayleigh profile geoid separation $Ray\ geoid_{profile}$:** Derived in the same way as for the corresponding Mie parameter:

$$Ray\ geoid_{profile}(p) = Ray\ geoid_{wr}(p_id_{Mie}(p, i_{max}))$$

where p describes the profile index, i_{max} the range gate index of the last (lowest in terms of altitude) valid Rayleigh wind result, $Ray\ geoid_{wr}$ the geoid separation values per wind result and p_id_{Ray} the profile-wind-id matrix for the Rayleigh channel.

Table 20: Parameters of [RD3] corresponding to formula variables of Section 3.3

Variable	VDDA-PT name
$Mie\ Lon_{DEM,wr}$	Mie wind result lon. of DEM intersection
$Mie\ Lon_{DEM,profile}$	Mie profile lon. of DEM intersection
$Mie\ Lat_{DEM,wr}$	Mie wind result lat. of DEM intersection
$Mie\ Lat_{DEM,profile}$	Mie profile lat. of DEM intersection
$Mie\ Alt_{DEM,wr}$	Mie wind result alt. of DEM intersection
$Mie\ Alt_{DEM,profile}$	Mie profile alt. of DEM intersection
$Mie\ geoid_{wr}$	Mie wind result geoid separation
$Mie\ geoid_{profile}$	Mie profile geoid separation
$Ray\ Lon_{DEM,wr}$	Rayleigh wind result lon. of DEM intersection
$Ray\ Lon_{DEM,profile}$	Rayleigh profile lon. of DEM intersection
$Ray\ Lat_{DEM,wr}$	Rayleigh wind result lat. of DEM intersection
$Ray\ Lat_{DEM,profile}$	Rayleigh profile lat. of DEM intersection
$Ray\ Alt_{DEM,wr}$	Rayleigh wind result alt. of DEM intersection
$Ray\ Alt_{DEM,profile}$	Rayleigh profile alt. of DEM intersection
$Ray\ geoid_{wr}$	Rayleigh wind result geoid separation
$Ray\ geoid_{profile}$	Rayleigh profile geoid separation
p_id_{Mie}	Mie wind profile wind result id
p_id_{Ray}	Rayleigh wind profile wind result id

3.4 Level 2C

The L2C product contains parameters at the same horizontal resolutions as described above for the L2B data product.

This product contains HLOS wind speeds and horizontal wind vector components after the assimilation process of L2B HLOS winds at the ECMWF. The horizontal wind vector is thereby described by the u- and v-wind component. The u-component (also known as zonal

wind) is positive for a flow from west to east (westerly wind) and the v-component (also known as meridional wind) is positive for south to north flow (southerly wind). Given u- and v-wind components, it is possible to calculate horizontal wind speed and wind direction. Those parameters can be visualised when selecting the L2C product. Horizontal wind velocity V_{hor} is thereby calculated according to following equation:

$$V_{hor} = \sqrt{u^2 + v^2}$$

The wind direction refers to meteorological wind direction, i.e. the direction from which the wind is blowing and increasing clockwise from the North. To calculate the meteorological wind direction φ_{met} [°] from the u- and v-wind components, the inverse of the atan2-function, an advanced arctangent function, is used. In contrast to the arctangent-function, the atan2-function takes the signs of both vector components into account and places the angle in the correct quadrant leading to following equation for φ_{met} :

$$\varphi_{met} = \frac{180^\circ}{\pi} \cdot \text{atan2}(u, v) + 180^\circ$$

The calculation of the horizontal wind speed and wind direction is performed for the L2C analysis wind components as well as for the L2C background wind components. Thereby, following new parameters are created:

- **Mie assimilation background horizontal wind velocity $Mie\ wspeed_{assim,bkg}$:**

$$Mie\ wspeed_{assim,bkg}(w_{Mie}) = \sqrt{Mie\ u_{assim,bkg}^2 + Mie\ v_{assim,bkg}^2}$$

where w_{Mie} is the wind result index for the Mie channel and $Mie\ u_{assim,bkg}$ and $Mie\ v_{assim,bkg}$ are the assimilated background wind components u and v of the Mie channel.

- **Rayleigh assimilation background horizontal wind velocity $Ray\ wspeed_{assim,bkg}$:**

$$Ray\ wspeed_{assim,bkg}(w_{Ray}) = \sqrt{Ray\ u_{assim,bkg}^2 + Ray\ v_{assim,bkg}^2}$$

where w_{Ray} is the wind result index for the Rayleigh channel and $Ray\ u_{assim,bkg}$ and $Ray\ v_{assim,bkg}$ are the assimilated background wind components u and v of the Rayleigh channel.

- **Mie assimilation background wind direction $Mie\ wdir_{assim,bkg}$:**

$$Mie\ wdir_{assim,bkg}(w_{Mie}) = \frac{180^\circ}{\pi} \cdot atan2(Mie\ u_{assim,bkg}, Mie\ v_{assim,bkg}) + 180^\circ$$

where w_{Mie} is the wind result index for the Mie channel and $Mie\ u_{assim,bkg}$ and $Mie\ v_{assim,bkg}$ are the assimilated background wind components u and v of the Mie channel.

- **Rayleigh assimilation background wind direction $Ray\ wdir_{assim,bkg}$:**

$$Ray\ wdir_{assim,bkg}(w_{Ray}) = \frac{180^\circ}{\pi} \cdot atan2(Ray\ u_{assim,bkg}, Ray\ v_{assim,bkg}) + 180^\circ$$

where w_{Ray} is the wind result index for the Rayleigh channel and $Ray\ u_{assim,bkg}$ and $Ray\ v_{assim,bkg}$ are the assimilated background wind components u and v of the Rayleigh channel.

- **Mie assimilation analysis horizontal wind velocity $Mie\ wspeed_{assim,analysis}$:**

$$Mie\ wspeed_{assim,analysis}(w_{Mie}) = \sqrt{Mie\ u_{assim,analysis}^2 + Mie\ v_{assim,analysis}^2}$$

where (w_{Mie}) is the wind result index for the Mie channel and $Mie\ u_{assim,analysis}$ and $Mie\ v_{assim,analysis}$ are the assimilated analysis wind components u and v of the Mie channel.

- **Rayleigh assimilation analysis horizontal wind velocity $Ray\ wspeed_{assim,analysis}$:**

$$Ray\ wspeed_{assim,analysis}(w_{Ray}) = \sqrt{Ray\ u_{assim,analysis}^2 + Ray\ v_{assim,analysis}^2}$$

where w_{Ray} is the wind result index for the Rayleigh channel and $Ray\ u_{assim,analysis}$ and $Ray\ v_{assim,analysis}$ are the assimilated analysis wind components u and v of the Rayleigh channel.

- **Mie assimilation analysis wind direction $Mie\ wdir_{assim,analysis}$:**

$$Mie\ wdir_{assim,analysis}(w_{Mie}) = \frac{180^\circ}{\pi} \cdot atan2(Mie\ u_{assim,analysis}, Mie\ v_{assim,analysis}) + 180^\circ$$

where w_{Mie} is the wind result index for the Mie channel and $Mie u_{assim,analysis}$ and $Mie v_{assim,analysis}$ are the assimilated analysis wind components u and v of the Mie channel.

- **Rayleigh assimilation analysis wind direction $Ray wdir_{assim,analysis}$:**

$$Ray wdir_{assim,analysis}(w_{Ray}) = \frac{180^\circ}{\pi} \cdot atan2(Ray u_{assim,analysis}, Ray v_{assim,analysis}) + 180^\circ$$

where w_{Ray} is the wind result index for the Rayleigh channel and $Ray u_{assim,analysis}$ and $Ray v_{assim,analysis}$ are the assimilated analysis wind components u and v of the Rayleigh channel.

Table 21: Parameters of [RD3] corresponding to formula variables of Section 3.4

Variable	VDDA-PT name
$Mie wspeed_{assim,bkg}$	Mie assimilation background horizontal wind velocity
$Mie u_{assim,bkg}$	Mie assimilation background u wind velocity
$Mie v_{assim,bkg}$	Mie assimilation background v wind velocity
$Ray wspeed_{assim,bkg}$	Rayleigh assimilation background horizontal wind velocity
$Ray u_{assim,bkg}$	Rayleigh assimilation background u wind velocity
$Ray v_{assim,bkg}$	Rayleigh assimilation background v wind velocity
$Mie wdir_{assim,bkg}$	Mie assimilation background wind direction
$Ray wdir_{assim,bkg}$	Rayleigh assimilation background wind direction
$Mie wspeed_{assim,analysis}$	Mie assimilation analysis horizontal wind velocity
$Mie u_{assim,analysis}$	Mie assimilation analysis u wind velocity
$Mie v_{assim,analysis}$	Mie assimilation analysis v wind velocity
$Ray wspeed_{assim,analysis}$	Rayleigh assimilation analysis horizontal wind velocity
$Ray u_{assim,analysis}$	Rayleigh assimilation analysis u wind velocity
$Ray v_{assim,analysis}$	Rayleigh assimilation analysis v wind velocity
$Mie wdir_{assim,analysis}$	Mie assimilation analysis wind direction

<i>Ray wdir_{assim,analysis}</i>	Rayleigh assimilation analysis wind direction
--	---

3.5 Auxiliary RRC

Parameters of this data product are specified per calibration frequency step. For the calculation of one frequency step a certain number of observations is used. The only parameter of this product specified per observation is the geoid separation value. In order to change the altitude references of the range bin altitudes which are given per frequency step, a new parameter needs to be introduced which assigns one geoid separation value to each frequency step location:

- **Geoid separation freq. step N_{RRC} :** This parameter is derived by calculating the mean of the geoid separation values $N_{RRC,obs}$ contributing to the frequency step:

$$N_{RRC}(f) = \frac{1}{J} \sum_{j=1}^J N_{RRC,obs}(f, j)$$

where f denotes the frequency step index, j the observation index and J the number of observations contributing to one frequency step.

Table 22: Parameters of [RD3] corresponding to formula variables of Section 3.5

Variable	VDDA-PT name
$N_{RRC,obs}$	Geoid separation obs.

4 Algorithms

The structure of this Section is based on the operational scenarios “Visualisation”, “Filtering”, “Model and Product Analysis” and “Download and Saving”.

4.1 Concatenation of multiple data sets

Data downlink from the satellite to the ground station in Svalbard (Norway) is foreseen for each visible orbit. Therefore, Aeolus data product files usually contain data from one orbit. For the case that orbits are not visible for the ground station in Svalbard (blind orbits), product files containing data from multiple orbits (~2-5 orbits) are downlinked with the next possible overpass at Svalbard. Due to the downlink strategy i.e. to preserve data generation in near real time, an overlap between data product files from two successive orbits exists. This overlap is not corrected throughout the processing from AISP to L0 to L1B and therefore exists for all Aeolus data product types at all levels.

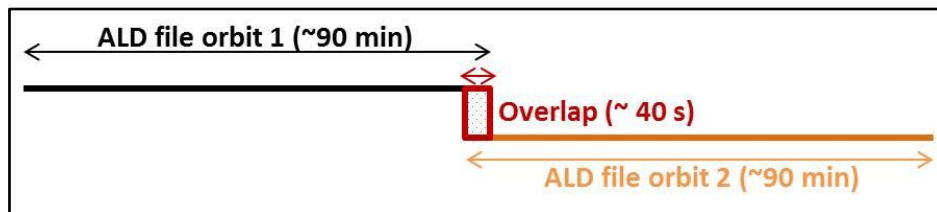


Figure 1: Data overlap between two consecutive orbits

In order to import and process multiple data product files, a specification for the concatenation of the datasets is necessary (see UR-VIR_AE-FUN_02040 in [RD1]). Figure 1 illustrates two adjacent orbits containing ALADIN data and the overlap between two orbits. One orbit takes about the 90 minutes and the overlap between two data product files usually extends over 40 seconds (Fabio Buscaglione (ESA-ESRIN), email-correspondence (2017-08-25)). For the case that the selected ToI spans over two orbits, the concatenated file would contain identical measurements and observations. For data product types on measurement and observational level (L1B, L2A), the procedure to remove identical measurements and observations is straightforward. The measurement times and observation times (parameters given in [RD3]) of the data product types are scanned for duplicates. The thereby detected indices are used to remove duplicate data for all other datasets (product confidence, measurement ...) from the concatenated data product. For L1B auxiliary data products, i.e. AUX_ISR, AUX_MRC, AUX_RRC and AUX_ZWC, it cannot happen that data is split between two files since preliminary merging of the concerned files is performed before submission to the L1B processor (Fabio Buscaglione (ESA-ESRIN) pers.com.).

For the L2B and L2C data product types which are based on the grouping of measurements and the accumulation of measurements to wind results, the removal of duplicates is different. Figure 2 shows grouped measurement data from two successive orbits. Due to difference in availability of L1B measurement data, the grouping at the end of the L2B/L2C orbit 1 and at the beginning of L2B/L2C orbit 2 and consequently the accumulated wind results can be different. The first group of the second orbit incorporates more information than the last group of the first orbit. Thus, only data **from the second orbit** is used for the overlap areas since the grouping of the second orbit is not abruptly interrupted by the end of file [RD12]. Thus, parameters from the second orbit are considered to be more reliable than results from the first orbit. Furthermore, it has to be taken into account that the L2B/L2C product contains data sets with different granularities (measurement-, group-, wind result- and profile-level). Hence, L2B/L2C data sets of different granularities are scanned separately for duplicate time stamps to identify the overlap areas and measurement data from the first L2B/L2C file is removed. Since groups of the L2A data product are only defined within one single observation, duplicates can be removed applying the same procedure as for the L1B data product described above.

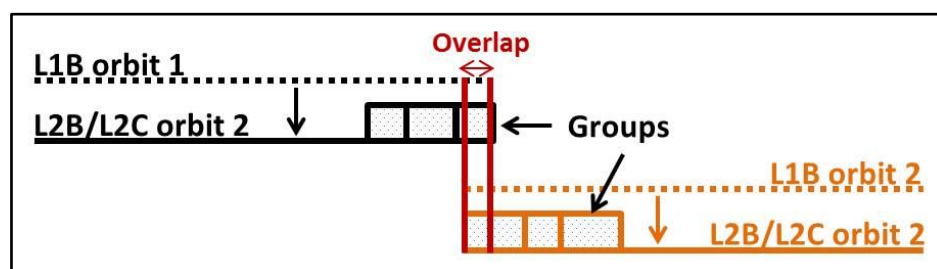


Figure 2: L2B/L2C overlap between two consecutive orbits

4.2 Aeolus surface albedo map from ADAM

Good knowledge of the expected surface reflectivity is of importance for many optical satellite-based Earth observation missions. For Aeolus, additional information about the surface reflectance is an important input for the characterisation of ground measurements since the accuracy of ground calibrations is influenced by the surface albedo. Thus, the generation and visualisation of surface albedo maps is a central part of VirES for Aeolus.

For this purpose, surface reflectance maps are created using products of version 3 of the ADAM (**A** Surface Reflectance **D**atabase for ESA's Earth Observations **M**issions) database. The ADAM data base is made up of datasets of gridded climatologies (representative of the year 2005) of normalised MODIS reflectance values over land, chlorophyll concentration and wind speed over ocean. ADAM allows the computation of narrow-band surface reflectances over the spectral domain ranging from 240-4000 nm at 1 nm spectral resolution in any observation geometry with a spatial resolution of $0.1^\circ \times 0.1^\circ$. Thereby, specific processing depending on the surface type is performed as surface reflectance characteristics of different surface types are taken into account. Data gaps for pixels, mainly existent for ocean and land pixels affected by sea-ice coverage, are filled applying gap-filling procedures [WR1] [WR2].

An Application Programming Interface (API) written in the Python programming language has been made available allowing the users to download the entire ADAM data base and perform operations locally on their hardware according to their specific requirements [WR3]. The Python code is installed and used in the same way as Python modules. In order to derive surface albedo maps for Aeolus requirements, the tools of API are used. Pseudo-code including comments of the Python script used for the calculations to derive the surface albedo values for Aeolus is shown in Figure 4.

For Aeolus, monthly global surface albedo maps at a wavelength of 355 nm and for off-nadir and nadir observation geometry are created. For the off-nadir reflectance maps, surface reflectance is computed using an illumination- and viewing-geometry of 37.6° for direct backscatter. To generate reflectance maps representative for the satellite's nadir pointing, the illumination and viewing angles are set to zero degrees. In this way, 24 global surface reflectance maps, two (nadir and off-nadir) representative for each month, with a spatial resolution of $0.1^\circ \times 0.1^\circ$ (11 km x 11 km at the equator) are generated. Due to the fact that the ADAM gap-filling procedure for sea-ice pixels does not account for the spatial variation of the sea-ice coverage, it is considered unsatisfactory for Aeolus. Therefore, post-processing of the generated maps is necessary. For the ADAM gap-filling, sea ice pixels are already identified using sea-ice maps of the CryoClim database (osisaf-nh and osisaf-sh) which provide monthly climatologies of sea-ice concentration for the polar regions of the northern and southern hemisphere representative for the year 2005 [WR6]. For each sea-ice pixel, gap-filling is performed by attributing the averaged reflectance of the ten closest snow/ice pixels. This approach results in large areas with constant sea ice reflectance values where the spatial variability of the sea-ice coverage is captured insufficiently. For the post-processing step, the information about the sea-ice concentration per pixel ice_{conc} provided by the CryoClim sea-ice maps is used to overcome this problem. The value ice_{conc} defines the area of sea ice relative to the total area of the pixel in percent. The reflectance value of sea-ice pixels is considered to be a linear combination of the reflectance of ice and water which leads to following equation for the reflectance of sea-ice pixels $R_{sea\ ice}$:

$$R_{sea\ ice} = \frac{ice_{conc}}{100} \cdot \bar{R}_{snow,ice} + \left(1 - \frac{ice_{conc}}{100}\right) \cdot \bar{R}_{ocean}$$

where $\bar{R}_{snow,ice}$ and $\bar{R}_{ocean}$ describe mean values for snow/ice and ocean pixels, respectively. The mean reflectance values vary each month and are different for the northern and the southern hemisphere. For the correction of sea-ice pixels in the northern hemisphere, $\bar{R}_{snow,ice}$ is derived from snow/ice pixels located in Greenland and $\bar{R}_{ocean}$ from ocean pixels in the Arctic. Mean values for snow/ice and ocean representative for the southern hemisphere are derived using snow/ice pixels from the Antarctica and ocean pixels close to the Antarctica. Figure 3 shows global maps of surface reflectances derived for an illumination angle and viewing angle of 37.6° for backscatter at a wavelength of 355 nm for the month July. The top plot shows the reflectance maps where no post-processing for the sea-ice pixels is performed. The lack of spatial variability in the sea ice reflectance is clearly visible. The bottom plot illustrates the same map after post-processing of the sea-ice reflectances under consideration of the spatial variability of the sea-ice coverage according to the equation above.

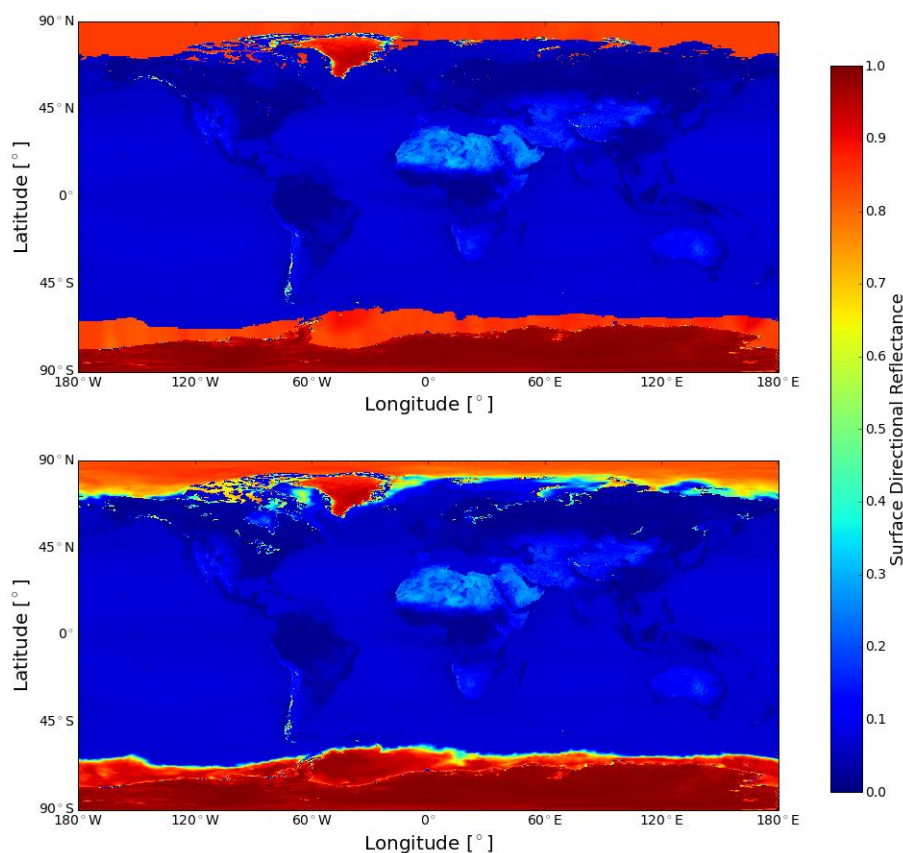


Figure 3: ADAM surface reflectance maps calculated for off-nadir viewing geometry representative for the month of July. Top/Bottom: before/after post-processing of sea-ice pixels.

```
import adam
import adam_config
import adam.process_brdf
import adam.process_reflectance

# loading of ADAM modules

-----
my_config = adam_config.adam_config(path_data = data_sourcedir, output_root_dir = data_savedir)

# The API needs to be initialized with a configuration file specifying the location of the downloaded ADAM data
base (path data) and a location where to write output data (output_root_dir).

request_dict = { 'fieldOperationType' : 'brdf',      # Perform a BRDF calculation
                 'fieldMonth'         : 'month',     # Three first letters of the month
                 'fieldSunZenith'      : 37.6 / 0,    # Illumination angle (37.6° for off-nadir, 0° for nadir)
                 'fieldViewZenith'     : 37.6 / 0,    # Viewing angle (37.6° for off-nadir, 0° for nadir)
                 'fieldRelAzimuth'     : 0,          # Rel. azimuth angle (0° refers to direct backscatter)
                 'fieldSpectralDomain' : '355-355',   # Spectral domain of interest in nm (355 nm for Aeolus)

                 'fieldLatMin'         : lat_min,     # minimum value of the latitude [-90°,90°]
                 'fieldLonMin'         : lon_min,     # minimum value of the longitude [-180°,180°]
                 'fieldLatMax'         : lat_max,     # maximum value of the latitude [-90°,90°]
                 'fieldLonMax'         : lon_max,     # maximum value of the longitude [-180°,180°]
                 }

## In this section the input parameters for the following calculations are specified
-----
job = adam_tools.adam.adam_job(cfg = my_config)  # Initialization of the job according to "my_config"

job.validate_input(request_dict)                # Pass the input parameters

job.load_data()                                # load raw data into the job

job.process_reflectance()                       # Calculate surface reflectance

job.process_brdf()                             # Perform BRDF calculation

## Here, the reflectance calculations are performed according to the inputs specified above.
-----
job.data['SDR']

## In this way, the spectro-directional reflectance in the user defined observation geometry and in the output
spectral band is accessed.
-----
```

Figure 4: Pseudo Python-code of the API showing the steps needed to derive surface reflectance data for Aeolus requirements including comments (red).

4.3 Display of surface albedo values

For the proper visualisation of the generated surface reflectance maps in the Globe View, several steps are necessary. The albedo map suitable for the selected scenario depending on the instrument mode and the ToI has to be chosen. For the case that the data products AUX_MRC and AUX_RRC where the satellite is operated in nadir mode are shown, the surface reflectance maps generated at nadir viewing geometry are selected. For all other data products maps at off-nadir geometry are visualised. The ToI determines which of the twelve surface albedo maps is plotted. If the user-selected time period lies within the range of one month, the reflectance map of the corresponding map is visualised. For the case that the ToI extends over two months, the number of days of the ToI in the two months defines the selection of the reflectance map. The month including more days of the ToI determines the selection of the albedo map. Given that the ToI spans from the 30th of July to the 4th of August, the reflectance map of the month August is chosen for visualisation by default. Additionally, it is possible for the user to select between the reflectance maps of the corresponding months.

For the Analytics View, it is possible to visualise surface albedo values at the LOS intersections with the DEM. Therefore, temporal allocation of the right map according to the selected ToI is necessary. This is problematic when the selected ToI spans over two months. Since the surface albedo maps are based on monthly climatologies, non-physical gradients in the surface albedo would occur at the transition of the two months. Therefore, the same procedure as described above for the Globe View is applied which only assigns one albedo map according to the number of days in the corresponding months. In a next step, spatial allocation between the LOS intersections with the DEM of the Aeolus data products is performed. Therefore, each geolocation value of the ALADIN data product is assigned to an albedo value. This is done by selecting the nearest neighbour of the geolocation (latitude and longitude) of the intersection point with the DEM on the surface albedo map. Through that, it is also possible to filter the ALADIN data products according to these albedo values at the intersection points.

4.4 Display of multiple y-axes

Altitude parameters of the different ALADIN data products are based on different reference altitudes (summarised in Table 23). For instance, altitudes of the L1B data product are given w.r.t the WGS 84 (World Geodetic Model 1984) ellipsoid, whereas range gate altitudes of both channels of the L2B product are referenced to the geoid model EGM 96 (Earth Gravitational Model 1996). The WGS 84 reference ellipsoid is an idealised representation of the elliptical shape of the Earth's surface. In contrast to that, the geoid model EGM 96 is a geopotential model of the earth taking into account slight variations in the acceleration of gravity due to variations in size, shape and mass distribution of the earth. Since the sea level is determined by the earth's gravitational field, altitudes referenced to the EGM96 Model are a measure of the mean sea level at a given point [WR8][WR9].

Figure 5 shows an Aeolus measurement section with the corresponding altitudes of DEM intersection referenced to the ellipsoid (h) and referenced to the geoid (H). The geoid separation term N which describes the height of the geoid above the ellipsoid is also shown in the figure. This separation can make up to 100 m depending on the geolocation. For the comparison and visualisation of altitude based parameters from different data products, a common altitude reference is of importance. Thus, it is considered the best way to give users the possibility to select between the reference levels (w.r.t ellipsoid or w.r.t geoid) when visualising altitude based parameters. In consequence, rules for the conversion between the two reference levels need to be specified.

The relationship between two the altitude reference levels can be expressed as follows:

$$H = h - N$$

where H denotes altitude w.r.t the geoid, h altitude w.r.t the ellipsoid and N the geoid separation term. N is included in the data product files for L1B, L2A, L2B and L2C and is used for the conversion between the two reference levels according to the formula described above. In the following sections the conversion between the two reference levels is explained for the different Aeolus data products:

- **L1B:** All altitude parameters of the L1B data product are referenced to the WGS 84 ellipsoid. For each observation the geoid separation parameter N is included in the data product. For the case that the user wants to change the reference level of the

different altitudes on observation level, the conversion from the ellipsoid to the geoid reference is performed according to following formulas:

$$\begin{aligned} Mie_{alt\ geoid,obs}(j,i) &= Mie_{alt,obs}(j,i) - N_{L1B}(j) \\ Ray_{alt\ geoid,obs}(j,i) &= Ray_{alt,obs}(j,i) - N_{L1B}(j) \\ Alt_{DEM\ geoid,obs}(j) &= Alt_{DEM,obs}(j) - N_{L1B}(j) \end{aligned}$$

where j is the observation index, i the range gate index, $Alt_{DEM,obs}$ the altitude of the DEM intersection on observation level and $Mie_{alt,obs}$ and $Ray_{alt,obs}$ denote the altitudes for each range bin for the Mie and Rayleigh channel, respectively.

For L1B parameters on measurement level, the geoid separation term valid for the corresponding observation is used:

$$\begin{aligned} Mie_{alt\ geoid,meas}(j,k,i) &= Mie_{alt,meas}(j,k,i) - N_{L1B}(j) \\ Ray_{alt\ geoid,meas}(j,k,i) &= Ray_{alt,meas}(j,k,i) - N_{L1B}(j) \\ Alt_{DEM\ geoid,meas}(j,k) &= Alt_{DEM,meas}(j,k) - N_{L1B}(j) \end{aligned}$$

where j is the observation index, k the measurement index, i the range gate index, $Alt_{DEM,meas}$ the altitude of the DEM intersection on measurement level and $Mie_{alt,meas}$ and $Ray_{alt,meas}$ denote the altitudes for each range bin for the Mie and Rayleigh channel, respectively.

- **L2A:** As for L1B parameters, L2A altitude parameters on observation level are also referenced to the EGM 96 geoid. Thus, the same equations as above are used for the conversion of the Mie and Rayleigh bin altitudes and for the altitude of the DEM intersection. The reference level of the altitude parameter describing the bin middle altitude for SCA middle-bin-parameters is changed as follows:

$$SCA_{alt\ middle\ bin\ ellipsoid,obs}(j,i) = SCA_{alt\ middle\ bin,obs}(j,i) + N_{L2A}(j)$$

where j is the observation index and i the range gate index and $SCA_{alt\ middle\ bin,obs}$ the bin middle altitudes (w.r.t ellipsoid) for SCA parameters.

In order to change the altitude reference for L2A group parameters the geoid separation term at the starting point the group is used to convert the altitude reference of the group:

$$Group_{alt\ ellipsoid}(g) = Ray_{alt,meas}(j = g_{obs,start}(g), k = g_{meas,start}(g), i = h_{group}(g)) + N_{L2A}(j = g_{obs,start}(g))$$

where g denotes the group index, j the observation index, k the measurement index and i the range gate index. The parameter $g_{obs,start}$ is the index of the first observation of the group, $g_{meas,start}$ the first measurement of the group and h_{group} is the index of the height level of the group in the Rayleigh scale.

The references of group middle-bin start and end altitude parameters are converted as follows:

$$\begin{aligned} Group_{middle\ bin\ start\ alt\ ellipsoid}(g) &= Group_{middle\ bin\ start\ alt}(g) + N_{L2A}(j) \\ &= g_{obs,start}(g) \\ Group_{middle\ bin\ stop\ alt\ ellipsoid}(g) &= Group_{middle\ bin\ stop\ alt}(g) + N_{L2A}(j) \\ &= g_{obs,stop}(g) \end{aligned}$$

where g is the group index, j the observation index, $g_{obs,start}$ the index of the first observation of the group and $Group_{middle\ bin\ start\ alt}$ and $Group_{middle\ bin\ stop\ alt}$ denote the start and stop altitudes of the group.

- **L2B/L2C:** L2B and L2C altitude parameters of the wind results as well as altitudes of the DEM intersection are given w.r.t the geoid model EGM 96. The conversion of the reference level from geoid to ellipsoid is performed according to following equations:

$$\begin{aligned} Mie_{alt\ ellipsoid}(w_{Mie}) &= Mie_{alt}(w_{Mie}) + N_{Mie,L2B/L2C}(w_{Mie}) \\ Ray_{alt\ ellipsoid}(w_{Ray}) &= Ray_{alt}(w_{Ray}) + N_{Ray,L2B/L2C}(w_{Ray}) \end{aligned}$$

where w_{Mie} and w_{Ray} are the indices of the accumulated wind results of the Mie and Rayleigh channel and Mie_{alt} and Ray_{alt} are altitude parameters (could be bottom-, CoG- and top altitude) of the two channels. The parameters $N_{Mie,L2B/L2C}$ and $N_{Ray,L2B/L2C}$ are the geoid separation terms for the Mie and Rayleigh channel, respectively.

In a similar way, the reference levels of the altitudes of the DEM intersection are converted:

$$\begin{aligned} Mie_{alt\ DEM\ ellipsoid}(w_{Mie}) &= Mie_{alt\ DEM}(w_{Mie}) + N_{Mie,L2B/L2C}(w_{Mie}) \\ Ray_{alt\ DEM\ ellipsoid}(w_{Ray}) &= Ray_{alt\ DEM}(w_{Ray}) + N_{Ray,L2B/L2C}(w_{Ray}) \end{aligned}$$

where w_{Mie} and w_{Ray} are the indices of the accumulated wind results of the Mie and Rayleigh channel and $Mie_{alt\ DEM}$ and $Ray_{alt\ DEM}$ are altitude of the DEM intersection for the two channels. The parameters $N_{Mie,L2B/L2C}$ and $N_{Ray,L2B/L2C}$ are the geoid separation terms for the Mie and Rayleigh channel, respectively.

- **AUX_RRC:** Altitude parameters of this data product are referenced w.r.t WGS 84 ellipsoid. The conversion of the altitude reference level from ellipsoid to geoid is performed according to following equations:

$$RRC_{alt\ geoid,obs}(j,i) = RRC_{alt,obs}(j,i) - N_{RRC}(j)$$

where j is the observation index, i the range gate index, $RRC_{alt,obs}$ the altitude values of the range bins on observation level and N_{RRC} the geoid separation parameter per frequency step.

Altitude parameters of the data products AUX_MRC and AUX_ZWC are given w.r.t to the WGS 84 ellipsoid but there is no geoid separation value included in the data products. Thus, no conversion to the geoid reference level is possible without the use of additional data (e.g. other data product files or EGM 96 correction data). AUX_MET altitude parameters are referenced to the EGM 96 geoid and there is also no geoid separation parameter included which would allow selecting between the two altitude reference levels.

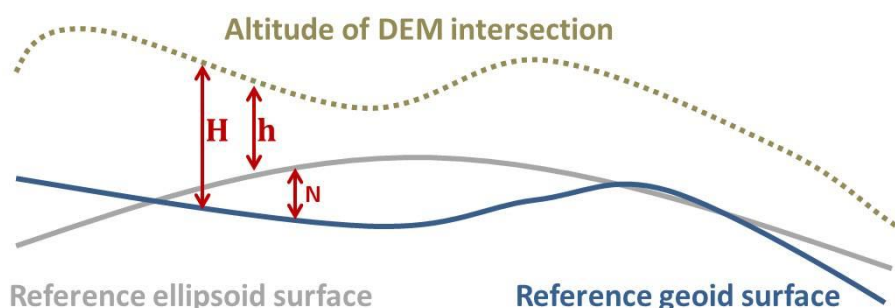


Figure 5: A sketch illustrating the relation between the ellipsoid altitude (h), the geoid altitude (H) of the DEM intersection and the geoid separation (N).

Table 23: Reference altitudes of different altitude parameters of different data products

	WGS 84 ellipsoid	EGM 96 geoid
L1B:		
Range gate altitude	x	
Alt. of DEM intersection	x	
L2A:		
Range gate altitude		x
Alt. of DEM intersection		x
L2B:		
Range gate altitude		x
Alt. of DEM intersection		x
L2C:		
Range gate altitude		x
Alt. of DEM intersection		x
AUX_MET:		
Surface altitude		x
Layer altitude		x
AUX_MRC:		

Range gate altitude	x	
AUX_RRC:		
Range gate altitude	x	
AUX_ZWC:		
Alt. of DEM intersection	x	

Table 24: Parameters of [RD3] corresponding to formula variables of Section 4.4

Variable	VDDA-PT name
L1B:	
$Mie_{alt,obs}$	Mie altitude obs.
$Ray_{alt,obs}$	Rayleigh altitude obs.
$Alt_{DEM,obs}$	Altitude of DEM intersection obs.
$Mie_{alt,meas}$	Mie altitude meas.
$Ray_{alt,meas}$	Rayleigh altitude meas.
$Alt_{DEM,meas}$	Altitude of DEM intersection meas.
N_{L1B}	Geoid separation obs.
L2A:	
$SCA_{alt\ middle\ bin,obs}$	SCA middle bin altitude obs.
N_{L2A}	Geoid separation obs.
L2B/L2C:	
Mie_{alt}	Mie wind result bottom/COG/top altitude
Ray_{alt}	Rayleigh wind result bottom/COG/top altitude
$Mie_{alt\ DEM}$	Mie wind result alt. of DEM intersection
$Ray_{alt\ DEM}$	Rayleigh wind result alt. of DEM intersection
$N_{Mie,L2B/L2C}$	Mie wind result geoid separation
$N_{Ray,L2B/L2C}$	Rayleigh wind result geoid separation
AUX_RRC:	
$RRC_{alt,obs}$	Altitude

N_{RRC}	Geoid separation freq. step
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Furthermore, it is common practice for meteorologists to use the pressure altitude as y-coordinate. It refers to the altitude corresponding to a given pressure in the International Standard Atmosphere (ISA). ISA is a conceptual model of the atmosphere describing the change of temperature, pressure, density and viscosity with altitude. For this, the air in the model is treated as perfect gas under the assumptions that the model air is devoid of dust and water vapour and that there are no winds or turbulence. The temperature within the model troposphere (until 11 km) follows a constant lapse rate of 6.5 K/1 000 m. The subsequent stratosphere is considered to be isothermal. The pressure altitude p is calculated using the hydrostatic equation, perfect gas law and taking into account the temperature lapse rate and can be expressed as a function of the altitude above mean sea level z (w.r.t geoid) as follows [WR7]:

$$p(z) = p_0 \cdot \left(1 - 0.0065 \frac{K}{m} \cdot \frac{z [m]}{T_0 [K]}\right)^{5.2561}$$

where p_0 and T_0 describe pressure and temperature at mean sea level (summarised in Table 25). This formula is valid for the troposphere ($z < 11000$ m). For the stratosphere ($z \geq 11000$ m) following expression is used:

$$p(z) = p_{11} \cdot e^{-\frac{g}{RT_{11}}(z-z_{11})}$$

where p_{11} , T_{11} , z_{11} denote pressure, temperature and altitude at 11 km and g and R are the acceleration of gravity and the specific gas constant for dry air, respectively (parameters summarised in Table 25). Figure 6 shows the pressure values calculated according the equations above as a function of the altitude. For the conversion of altitude parameters of the different data products into pressure altitudes, equations described above are used. This procedure is applied for every Aeolus data product with the exception of the AUX_MET product where values for the pressure altitude from the ECMWF model are already contained in the data product.

Table 25: Summary of parameters used for the calculation of the pressure altitude – parameters below are included in the VirES Aeolus Parameter file

Parameter	Meaning	Value
p_0	Pressure at mean sea level	1013.25 hPa
T_0	Temperature at mean sea level	288.15 K
p_{11}	Pressure at tropopause	226.32 hPa
T_{11}	Temperature at tropopause	216.65 K
z_{11}	Altitude at tropopause	11000 m

g	Acceleration of gravity	9.81 m s^{-2}
R	Specific gas constant for dry air	$287.06 \text{ J kg}^{-1}\text{K}^{-1}$

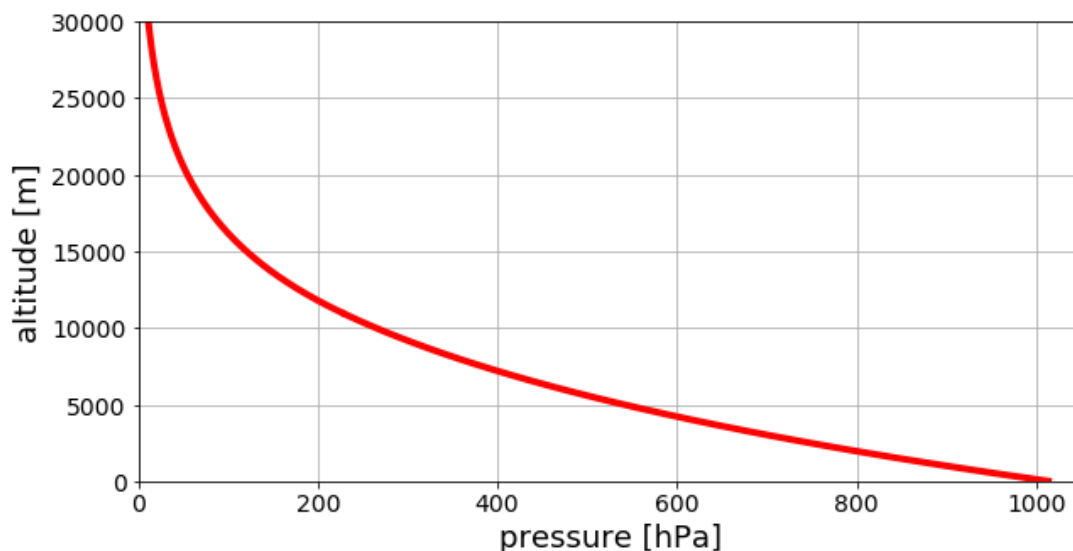


Figure 6: ISA-pressure (x-axis) vs. altitude (y-axis)

4.5 Spatio-temporal grid operations

In the Analytics View, it is possible to visualise parameters from different data products of different granularity. Moreover, there is the possibility to compare parameters from different data products by the visualisation as scatterplots (e.g. L1B winds vs. L2B winds), to filter with parameters from different data products and to calculate differences between parameters from different data products. For such operations, only certain combinations of different data products are eligible. It is possible to compare L2B accumulation results, L2C accumulation results, L2A grouped results and L2A observations with L1B parameters on observation level. Apart from that, the comparison between L2B accumulated results and L2A parameters on group and observation level is supported. For such cases, spatio-temporal grid operations are implemented which set both parameters to the same resolution. Table 26 provides a summary of possible combinations of different data products and specifies the granularity for which the co-location can be performed.

Figure 7 shall help to understand the comparison of accumulated L2B parameters with L1B parameters on observation level. The figure shows range bins of L1B wind observations (black boxes) and accumulated wind results (e.g. Mie cloudy) of the L2B data product (red boxes) covering several L1B measurements. The red dots indicate the centre of gravity-time of the L2B wind result. Depending on the grouping and accumulation of the L2B processor, the number of L1B measurements covered by a L2B wind result varies from a couple of measurements to several observations. For the co-location, each L2B wind result is assigned to the L1B observation closest to the centre of gravity of time of the L2B wind results. Therefore, the time stamps "Time obs." (centre time of observation) and "Mie/Rayleigh wind result COG time" (see sheets L1B and L2B of [RD3]) of both parameters are compared. For example, for the first wind result at the left lower end of Figure 7, the L1B observation highlighted by a green border would be assigned to this wind result.

For the allocation of L1B observations to L2A grouped results, the L1B observation containing the L2A group result (L2A groups are defined within one single observation) is selected. For this, the centroid time of the L2A groups (parameter "Group centroid time" in sheet L2A of [RD3]) and L1B observation times (parameter "Time obs." in sheet L1B of [RD3]) are used to assign each L2A group result to the L1B observation containing the L2A group parameter. The co-location of L1B observations and L2A observations is straightforward since both parameters are on the same grid. Therefore, each L2A observation is assigned to the corresponding L1B observation. For the case, that no L1B observation exists, no value is assigned.

For the co-location between L2B accumulated results and L2A group results, each L2B result is assigned the L2A group result closest to centre of gravity-time of the L2B wind result. Therefore, the time stamps "Mie/Rayleigh wind result COG time" and "Group centroid time" (see sheets L2A and L2B of [RD3]) are compared. In a similar way, the co-location between L2B accumulated results and L2A parameters on observation level is performed. Therefore, each L2B result is allocated to the L2A observation closest to the centre-of gravity time of the L2B accumulated result using the centroid time of the L2A observations (parameters "L1B centroid time obs." and "Mie/Rayleigh wind result COG time" of sheets L2A and L2B in [RD3]). For the comparison of L2B and L2C accumulated results, no co-location is necessary since both parameters are on the same grid which simplifies the calculation of differences between assimilated data and observations.

In this way, it is possible to visualise parameters of different data products on the same grid, filter with parameters from different data products and calculate differences between parameters of different data products.

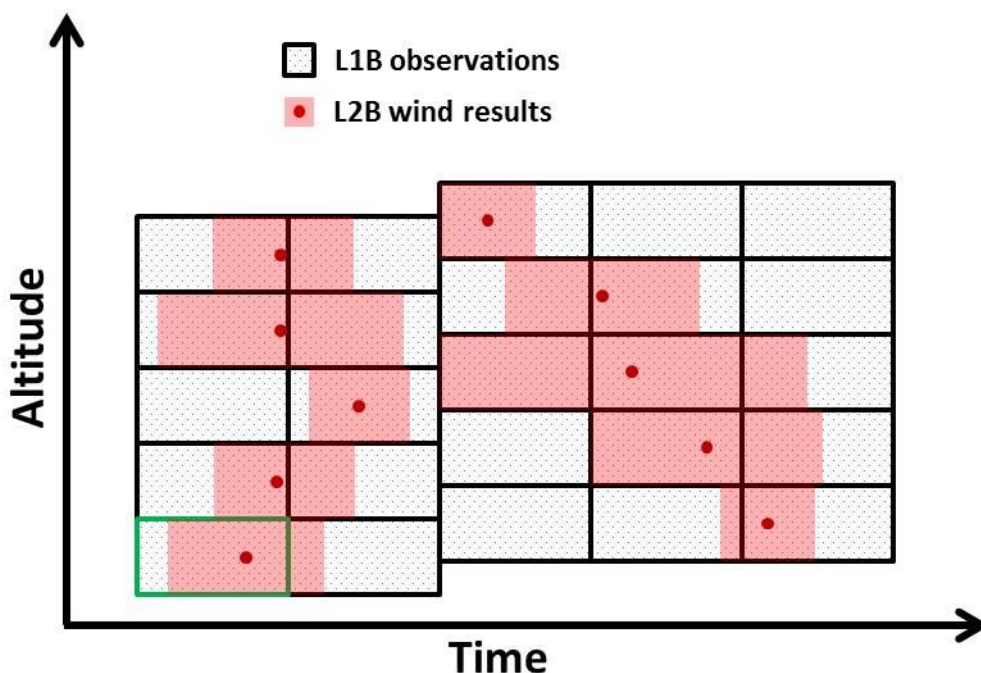


Figure 7: A sketch of L1B observations (black boxes) and L2B wind results (red boxes with centre of gravity).

Table 26: Summary of possible combinations for the co-location of different data products and the granularity at which the co-location is performed

	L1B	L2A	L2B
L2A	Group/obs. → obs.	-	-
L2B	Accumulation → obs.	Accumulation → groups/obs.	-
L2C	Accumulation → obs.	Accumulation → groups/obs.	Accumulation → Accumulation

4.6 Slice Widget

The Slice Widget allows the visualisation of 1D profiles or time series derived from horizontal and vertical slices of 2D data plots. It must be noted that the range bins of the different data product parameters are neither horizontally nor vertically aligned. Due to the possibility of terrain-following range bin setting, the range bin altitude/range can be different for the same range bin number. Moreover, considering the grouping algorithms of the L2A, L2B and L2C products, it generally occurs that the range bin size also differs on the horizontal scale. Therefore, a specification is needed that selects the right parameters for the 1D plot depending on the location of the range bins and the chosen positions of the horizontal and vertical slice lines.

Figure 8 shows a sketch of Aeolus range bins (e.g. L1B) as a function of time and altitude and how horizontal (blue) and vertical (red) slices could look like. The sketch demonstrates the assignment of range bins to the horizontal (blue bins) and vertical (red bins) slices. Such range bins that are transected by the horizontal or vertical slices are selected for the 1D plot. No further interpolation scheme is applied for cases when the range bins are transected at the bin top or bin bottom.

Figure 9 illustrates Aeolus range bins of the L2B data product as a function of time and altitude. Here, not also the horizontal alignment of the range bins changes due to the L2B accumulation algorithm. The same procedure as described above is applied in order to select range bins for the horizontal and vertical slices.

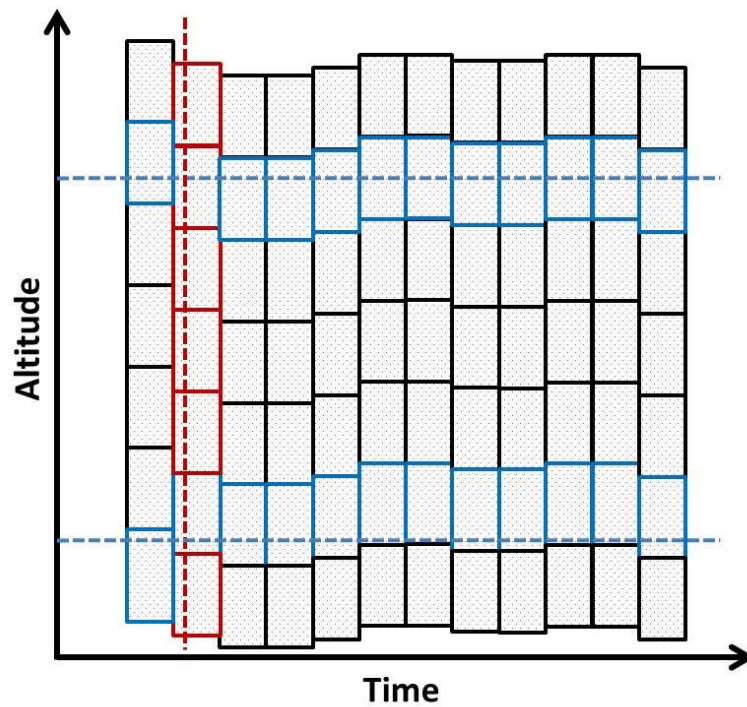


Figure 8: A sketch of Aeolus 2D range bins (boxes) of the L1B data product with horizontal (dashed blue line) and vertical (dashed red line) slices. The selected range bins selected for the horizontal and vertical slices are displayed as blue and red framed boxes, respectively.

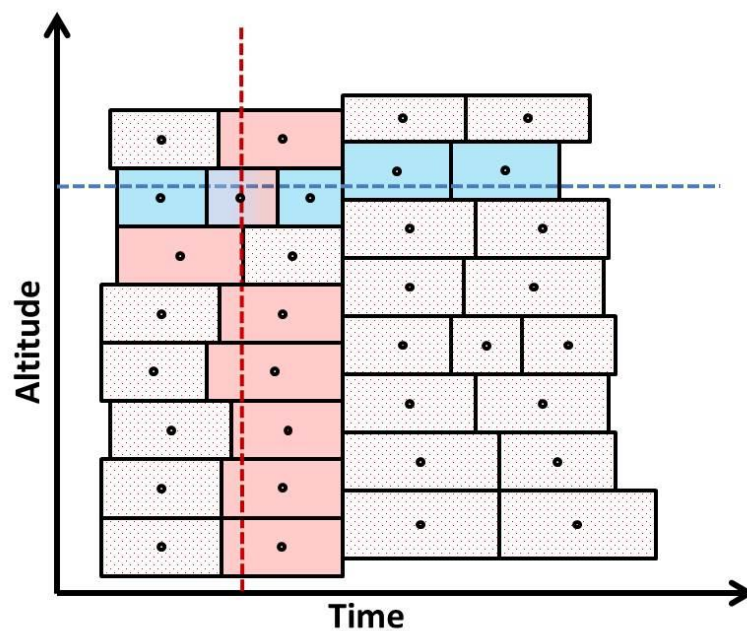


Figure 9: A sketch of Aeolus 2D range bins (boxes) of the L2B data product with horizontal (dashed blue line) and vertical (dashed red line) slices. The black dots refer to the CoG-times of the range bins. The selected range bins selected for the horizontal and vertical slices are displayed as blue and red boxes, respectively.

5 References, Definitions and Abbreviations

5.1 References

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- [WR7] International Standard Atmosphere:
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- [WR8] World Geodetic System 1984: <http://earth-info.nga.mil/GandG/publications/tr8350.2/wgs84fin.pdf>
- [WR9] Earth Gravitational Model 1996:
<https://cddis.nasa.gov/926/egm96/egm96.html>

5.2 Definitions

Baseline

Outcome of a distinct processing line based on defined parameters and/or algorithms to generate Data Products.

Dataset

A general term for data of any kind with no further specification about the data itself. A dataset is part of a data product or auxiliary or ancillary data. A dataset can contain several parameters.

Data Product / Data Product Types

The Term "Data Product" and "Data Product Type" refer to all processing levels (L0, L1A, L1B, L2A, L2B, L2C) of datasets measured by the ALADIN instrument and the Aeolus satellite platform or derived from them by processing. A "Data Product" is used for the different processing levels, e.g. L1B data product or L2B data product. Aeolus data product types are L0, L1A, L1B, L2A, L2B and L2C. A data product itself contains multiple datasets (e.g. geolocation, measurement, or product confidence dataset) and a dataset contains multiple parameters (e.g. latitude, time, wind speed) according to the IODD.

Auxiliary Data / Auxiliary Data Types

Represent additional data needed to process data products at various processing levels. Auxiliary data can be considered as another data product type. Auxiliary data contain multiple datasets and a dataset contains multiple parameters.

Calibration Data / Calibration Data Types

Data measured or observed by the Aeolus satellite platform which describes the performance, quality and calibration of the ALADIN instrument and the satellite platform and is required to process Data Products.

Ancillary Data / Ancillary Data Types

Data not derived or directly linked to Aeolus, but related in context and helpful for users.

Layer

A name used to select and handle data (any of the above) for visualisation and analysis in the VirES WebGUI. A layer is corresponding to a data product or auxiliary or ancillary data.

Operational Scenarios

The term "Operational Scenario" is defined as an imagined sequence of events that includes the interaction of the VirES Service with its environment and users, as well as interaction among service components. Operational Scenarios are grouped into the two categories System Operations (SO) and Service Utilization (SU). The latter group contains Operations such as "Filtering", "Download", "Product Analysis", or "Saving of Results".

Terms and Conditions

The following Terms and Conditions are distinguished:

- Terms and Conditions which govern the use of the VirES for Aeolus Service and its website, in short "**VirES Service Terms**" – specified by EOX;
- Terms and Conditions for the Utilisation of ESA's Earth Observation Data, which specifically govern the use of the Aeolus data available through VirES, in short "**ESA Data Terms**" – specified by ESA 2

Unclassified Information (as defined in ESA Security Directives – Annex 2, Ref. admin-ipol-2008-006e, 2008, par. 1.4.1):

"Unclassified information refers by default to any information, document or material, whatever the format or media, produced or acquired by ESA and which is not subject to formal security classification. As a general rule, unclassified information is not to be made publicly available, unless explicitly authorised."

Use Case

A set of actions or functions allowing the VirES System to achieve what is expected from the VirES Service. Stakeholders may utilize the Service for Use Cases such as "Quality

Analysis”, “Calibration and Validation”, “Scientific Exploitation”, or “Modelling and Prediction”. A specific Use Case is typically realized by combination of a number of Operational Scenarios.

VirES Service

Website and visual interface to the Aeolus mission products officially released by the European Space Agency. It includes tools to study atmospheric dynamics phenomena in particular those related to wind and Aerosols. It is structured in three areas or branches:

- Landing Page
- Workspace for discovery, view and analysis
- Download Manager

VirES Software

The software in scope is a bundle and configuration of:

- Software frameworks and libraries being developed and evolving largely independently from the VirES project (“software code dependencies”)
- Components being newly developed and evolving for the specific purposes of the VirES project (“VirES Software” or “VirES Application Software”)

VirES System

The system in scope is the union of VirES Software and the physical and virtual infrastructure on which it runs. This includes the VirES dedicated server- and user-side infrastructure. Within the system perimeter there are also code and configuration repositories as well as the System Operations and Administration Functions which are connected to the VirES dedicated server(s) and are hosted on provider platforms. The VirES System has well-defined data and control interfaces to surrounding systems.

5.3 Abbreviations

Table 27: List of Acronyms and Abbreviations

Acronym	Meaning
ACCD	Accumulation Charge Coupled Device
ACE2	Altimeter Corrected Elevation Version 2
ACMF	Aeolus Calibration and Monitoring Facility
AD	Applicable Document, Algorithm Design
ADAM	A surface reflectance Database for ESA’s Earth Observation Missions
ADM	Atmospheric Dynamics Mission
ADDF	Aeolus Data Dissemination Facility
AISP	Annotated Instrument Source Packets
ALADIN	Atmospheric LASer Doppler INstrument
APF	Aeolus Processing Facility
API	Application Programming Interface
AOCS	Attitude and Orbit Control System
AoI	Area of Interest
ATBD	Algorithm Theoretical Basis Document
a. u.	arbitrary unit
AUX	Auxiliary
CAL	Calibration
CALIPSO	Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observations
CAMS	Copernicus Atmosphere Monitoring Service
CCD	Charge Coupled Device

CLI	Command Line Interface
CoG	Centre-of-gravity
CPU	Central Processing Unit
CSR	Corrected Spectral Registration
CSV	Comma Separated Value
CZML	Cesium Modelling Language
D3	Data Driven Documents
DCMZ	Dark Current in Memory Zone
DEM	Digital Elevation Model
DLR	Deutsches Zentrum für Luft- und Raumfahrt, German Aerospace Center
DISC	Data Innovation and Science Cluster
DNS	Domain Network System
DoRIT	Dorit on Research and Information Technology
DS	Download and Saving
EarthCARE	Earth Clouds, Aerosols and Radiation Explorer
ECMWF	European Centre for Medium-Range Weather Forecasts
EGM 96	Earth Gravitational Model 1996
EO	Earth Observation
EOEP	Earth Observation Envelope Programme
ESRIN	European Space Research Institute
FAQ	Frequently Asked Questions
FOSS	Free and Open Source Software
FTP	File Transfer Protocol
GTS	Global Telecommunication System
GUI	Graphical User Interface
HBE	Harmonic Bias Estimator
HLOS	Horizontal Line-of-Sight
HTML	Hyper-Text Mark-up Language
HTTP	Hyper-Text Transfer Protocol
IAT	Instrument Auto Test
ICD	Interface Control Document
ICT	Information and Communication Technology
IODD	Input/Output Data Definitions
IRC	Instrument Response Calibration
ISA	International Standard Atmosphere
ISR	Instrument Spectral Registration
KNMI	Koninklijk Nederlands Meteorologisch Institute, Royal Netherlands Meteorological Institute
LI	Sign-up and Login
LOS	Line-of-Sight
MODIS	Moderate-resolution Imaging Spectroradiometer
MRC	Mie Response Calibration
(net)CDF	(Network) Common Data Format
NRT	Near-Real Time
NWP	Numerical Weather Prediction
OGC	Open Geospatial Consortium
OS	Operating System
OWS	Open Geospatial Consortium Web Services
PA	Product/Model Analysis
PCD	Product Confidence Data
PDGS	Payload Data Ground Segment
PT	Parameter Table
QWG	Quality Working Group

RAMS	Reliability, Availability, Maintainability and Safety
RBC	Rayleigh-Brillouin Correction
RD	Reference Document
RFQ	Request for Quotation
RRC	Rayleigh Response Calibration
RTD	Research and Technological Development
SI	Service Information
SO	System Operations
SS	System Specification
SSH	Secure Shell
SU	Service Utilization
SVG	Scalable Vector Graphics
TBD	To be Determined/Defined
TN	Technical Note
ToI	Time-period of Interest
URD	User Requirements Document
URL	Uniform Resource Locator
VDDA	VirES Data Dictionary Aeolus
VF	Visualization and Filtering
VirES	Virtual Workspaces for EO Scientists
VM	Virtual Machine
WebGL	Web Graphics Library
WGS 84	World Geodetic Model 1984
WMTS	Web Map Tile Service
w.r.t.	with respect to
WVM	Wind Velocity Mode
YUM	Yellowdog Updater, Modified
ZWC	Zero-Wind Calibration

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